Detailed Supplementary Calculations

for

A Modified Euler–Maruyama Method to Simulate a General One-Dimensional Sticky Diffusion

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This document contains expanded calculations underlying the proofs in the companion manuscript. It records intermediate integrations, substitutions, Taylor expansions, algebraic rearrangements, and moment calculations that are omitted from the manuscript and its concise Supplement for brevity.

Equation, theorem, lemma, assumption, and algorithm numbers refer to the companion manuscript version identified above. The manuscript and its concise Supplement contain the complete proof structure; this document is intended for readers who wish to inspect the intermediate calculations.

Companion manuscript: https://arxiv.org/abs/2606.27259

Appendix A. Supplement

A.1. Additional proofs from Section 3

Verification of (3.23) **in the proof of Lemma 3.5**

As a reminder, we wish to show that there exist constants $C_1(s, M) > 0$ and $\delta > 0$, independent of x and h, such that for all $h < \delta$,

$$Pe^{w(x)} \leq e^{w(x)} \left(1 - f_h(x) + C_1 h\right),$$

where f_h satisfies

$$f_h(x) \geq x\, \mathbf{1}_{\{x \in [0, \sqrt{3h}]\}}.$$

We would like to calculate $Pe^{w(x)} - e^{w(x)}$ for x in three cases: $[0, \frac{\sqrt{3h}}{2}]$, $[\frac{\sqrt{3h}}{2}, \sqrt{3h}]$ and $[\sqrt{3h}, +\infty)$.

(1) For $x \in [0, \frac{\sqrt{3h}}{2}]$, we compute

$$\begin{aligned}
Pe^{w(x)} &= \frac{\lambda(x)}{2} e^M + \frac{\lambda(x)}{\sqrt{3h}} \cdot \int_{\frac{\sqrt{3h}}{2}}^{\sqrt{3h}-x} e^{M - s\left(z - \frac{\sqrt{3h}}{2}\right)}\, dz + \frac{\lambda(x)}{2\sqrt{3h}} \cdot \int_{\sqrt{3h}-x}^{\sqrt{3h}+x} e^{M - s\left(z - \frac{\sqrt{3h}}{2}\right)}\, dz \\
&\quad + e^M(1 - \lambda(x)) \\
&= \lambda(x) \left(\frac{1}{2} e^M + e^M \cdot \frac{1}{2\sqrt{3h}s} \cdot \left(2 - e^{-s\left(\frac{\sqrt{3h}}{2} - x\right)} - e^{-s\left(\frac{\sqrt{3h}}{2} + x\right)} \right) \right) + e^M(1 - \lambda(x)).
\end{aligned}$$

In the first line, the first term corresponds to reflected jumps that land in the flat part of w, the two integral terms correspond to reflected jumps that land in the sloping part of w (split according to whether the reflection map has one or two preimages), and the last term corresponds to the jump directly to 0.

Now we simplify this expression by Taylor expansion of exponentials with a remainder of order $\mathcal{O}(h)$:

$$\begin{aligned}
&Pe^{w(x)} - e^{w(x)} \\
&= e^M \cdot \frac{\lambda(x)}{2\sqrt{3h}s} \cdot \left(2 - e^{-s\left(\frac{\sqrt{3h}}{2} - x\right)} - e^{-s\left(\frac{\sqrt{3h}}{2} + x\right)} \right) - e^M \cdot \frac{\lambda(x)}{2} \\
&= e^M \cdot \frac{\lambda(x)}{2\sqrt{3h}s} \cdot \left(\sqrt{3h}s - \frac{1}{2} s^2 \left(\frac{3h}{2} + 2x^2 \right) + \mathcal{O}(h^{3/2}) \right) - e^M \cdot \frac{\lambda(x)}{2} \\
&= e^M \cdot \frac{\lambda(x)}{2\sqrt{3h}s} \cdot \left(-\frac{1}{2} s^2 \left(\frac{3h}{2} + 2x^2 \right) + \mathcal{O}(h^{3/2}) \right) \\
&= e^M \cdot \left(-\frac{s}{4} \cdot \frac{\lambda(x)}{2} \sqrt{3h} - s \cdot \frac{\lambda(x) x^2}{2\sqrt{3h}} + \mathcal{O}(h) \right).
\end{aligned}$$

In other words, we have shown that, for $x \in [0, \sqrt{3h}/2]$,

$$Pe^{w(x)} \leq e^{w(x)} \left(1 - f_h(x) + \mathcal{O}(h)\right)$$

where we define

$$f_h(x) := \frac{s}{4} \cdot \frac{\lambda(x)}{2} \sqrt{3h} + s \cdot \frac{\lambda(x) \cdot x^2}{2\sqrt{3h}} \qquad \text{for } x \in [0, \frac{\sqrt{3h}}{2}]. \tag{A.1}$$

Recall the expression for $\mathbb{E}_x(\Delta X)$, (3.9) and substitute the expression for $\lambda(x)$, (3.3),

$$\begin{aligned}\mathbb{E}_x(\Delta X_k) &= -x + \frac{\lambda(x)\sqrt{3h}}{2} + \frac{\lambda(x)x^2}{2\sqrt{3h}} \\ &= \frac{1}{2\sqrt{3h}} \cdot \frac{\left(x-\sqrt{3h}\right)^2 (x^2+h)}{\left(\frac{\kappa}{\sqrt{3h}}+1\right)x^2 + h + \kappa\sqrt{3h}} \\ &\geq 0.\end{aligned}$$

Hence,

$$f_h(x) - x > \mathbb{E}_x(\Delta X) > 0, \quad \forall x \in [0, \frac{\sqrt{3h}}{2}].$$

(2) For $x \in [\frac{\sqrt{3h}}{2}, \sqrt{3h}]$, the scheme does the same jump, but the expression for $e^{w(x)}$ is different, which makes the calculation much more complex. It can be checked that

$$\begin{aligned}Pe^{w(x)} &= \lambda(x)\left(e^M \cdot \frac{\sqrt{3h}-x}{\sqrt{3h}} + e^M \cdot \frac{\frac{\sqrt{3h}}{2} - (\sqrt{3h}-x)}{2\sqrt{3h}}\right) \\ &\quad + \frac{\lambda(x)}{2\sqrt{3h}} \cdot \int_{\frac{\sqrt{3h}}{2}}^{\sqrt{3h}+x} e^{M-s\left(z-\frac{\sqrt{3h}}{2}\right)}\, dz + (1-\lambda(x))e^M \\ &= \lambda(x)e^M \cdot \frac{3\sqrt{3h} - 2sx + 2 - 2e^{-s\left(\frac{\sqrt{3h}}{2}+x\right)}}{4\sqrt{3h}s} + (1-\lambda(x))e^M.\end{aligned}$$

Since $e^{w(x)} = e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)}$,

$$Pe^{w(x)} - e^{w(x)} = A + B$$

where

$$\begin{aligned}A &= \lambda(x)e^M \cdot \frac{3\sqrt{3h} - 2sx + 2 - 2e^{-s\left(\frac{\sqrt{3h}}{2}+x\right)}}{4\sqrt{3h}s} - \lambda(x)\cdot e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)} \\ B &= (1-\lambda(x))\cdot e^M - (1-\lambda(x))\cdot e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)}.\end{aligned}$$

Simplifying A, B by expanding the exponential, we have

$$\begin{aligned}A &= \frac{\lambda(x)}{4\sqrt{3h}s} \cdot e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)} \cdot \left((3\sqrt{3h}s - 2sx + 2)e^{s\left(x-\frac{\sqrt{3h}}{2}\right)} - 2e^{-s\sqrt{3h}} - 4\sqrt{3h}s\right) \\ &= e^{w(x)} \cdot \frac{\lambda(x)}{4\sqrt{3h}s} \cdot \left(3\sqrt{3h}s^2x - \frac{9}{4}s^2 \cdot 3h - s^2x^2 + \mathcal{O}(h^{3/2})\right) \\ &= \frac{\lambda(x)}{4}e^{w(x)}\left(3sx - \frac{9}{4}\sqrt{3h}s - \frac{sx^2}{\sqrt{3h}} + \mathcal{O}(h)\right).\end{aligned} \tag{A.2}$$

Meanwhile,

$$
\begin{aligned}
B &= e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)} \cdot (1-\lambda(x)) \left(e^{s\left(x-\frac{\sqrt{3h}}{2}\right)} - 1\right) \\
&= e^{M-s\left(x-\frac{\sqrt{3h}}{2}\right)} \cdot (1-\lambda(x)) \left(s\left(x-\frac{\sqrt{3h}}{2}\right) + \mathcal{O}(h)\right).
\end{aligned} \tag{A.3}
$$

Combining (A.2), (A.3) together, we have

$$
Pe^{w(x)} - e^{w(x)} = A + B = -e^{w(x)} \left(f_h(x) + \mathcal{O}(h)\right)
$$

where we define

$$
f_h(x) := \frac{s\lambda(x)}{16}\sqrt{3h} + \frac{s\lambda(x)x^2}{4\sqrt{3h}} + \frac{s\lambda(x)x}{4} + \frac{s\sqrt{3h}}{2} - sx \quad \text{for } x \in [\frac{\sqrt{3h}}{2}, \sqrt{3h}]. \tag{A.4}
$$

It can be checked by derivative that $f_h(x)$ is decreasing for $x \in [\frac{\sqrt{3h}}{2}, \sqrt{3h}]$, and $f_h(\sqrt{3h}) = \frac{s}{16}\sqrt{3h} > \sqrt{3h}$, so we must have that

$$
f_h(x) - x > 0, \quad \forall x \in [\frac{\sqrt{3h}}{2}, \sqrt{3h}].
$$

(3) For $x \in (\sqrt{3h}, +\infty)$: in this region, the scheme does a standard EM jump. Using the fact that $\mathbb{E}_x(\Delta X) = 0$, and then using the fact that w is a piecewise linear function with a constant piece near the boundary, so it smaller than its linear part: $w(x+\Delta X) \leq M - s\left((x+\Delta X) - \frac{\sqrt{3h}}{2}\right)$, we have

$$
\begin{aligned}
Pe^{w(x)} &= \mathbb{E}_x\left(e^{w(x+\Delta X)}\right) \\
&\leq \mathbb{E}_x\left(e^{w(x)-s\cdot\Delta X}\right) \\
&= e^{w(x)} \cdot (1 - s\cdot\mathbb{E}_x(\Delta X) + \mathcal{O}(h)) \\
&= e^{w(x)} \cdot (1 + 0\cdot(-s) + \mathcal{O}(h)) \\
&= e^{w(x)} \left(1 - f_h(x) + \mathcal{O}(h)\right)
\end{aligned}
$$

where

$$
f_h(x) := 0, \qquad \text{for } x \in (\sqrt{3h}, +\infty). \tag{A.5}
$$

So far, we have shown that for $x \in [0, +\infty)$,

$$
Pe^{w(x)} \leq e^{w(x)} \left(1 - f_h(x) + \mathcal{O}(h)\right) \tag{A.6}
$$

where $f_h(x) \geq x \cdot \mathbf{1}_{x\in[0,\sqrt{3h}]}$ is defined according to (A.1),(A.4),(A.5), and the bounding constant of the big-$\mathcal{O}$ term depends on s. In each of the three cases above, the remainder term $\mathcal{O}(h)$ is uniform in x on the corresponding interval. Therefore, after taking the maximum of the three remainder constants and the minimum of the corresponding smallness thresholds, there exist constants $C_1(s, M) > 0$ and $\delta > 0$, independent of x and h, such that for all $h < \delta$,

$$
Pe^{w(x)} \leq e^{w(x)}\big(1 - f_h(x) + C_1 h\big), \qquad x \geq 0.
$$

Together with the bounds established above,

$$
f_h(x) \geq x\,\mathbf{1}_{\{x\in[0,\sqrt{3h}]\}},
$$

which proves (3.23).

A.2. **Additional proofs from Section 4**

Lemma 4.1. There exists $\delta > 0$ and constants $b_1, b_2, \sigma_1, \sigma_2$ such that, for every $h < \delta$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2],$$

whenever

$$x - \sigma(x)\sqrt{3h} < 0 \qquad \text{or} \qquad x - \sigma(x)\sqrt{3h} + b(x)h < 0.$$

Proof. We first show that the boundary layer is contained in a fixed compact neighborhood of 0 for all sufficiently small h. By Assumption 4, there exists $L > 0$ such that

$$|b(x)| + |\sigma(x)| \leq L(1 + x), \qquad x \geq 0.$$

If $x - \sigma(x)\sqrt{3h} < 0$, then

$$x < \sigma(x)\sqrt{3h} \leq |\sigma(x)|\sqrt{3h} \leq L(1 + x)\sqrt{3h}.$$

If $x - \sigma(x)\sqrt{3h} + b(x)h < 0$, then

$$x < \sigma(x)\sqrt{3h} - b(x)h \leq |\sigma(x)|\sqrt{3h} + |b(x)|h \leq L(1 + x)(\sqrt{3h} + h).$$

Hence, after choosing $\delta_0 > 0$ small enough so that

$$L(\sqrt{3\delta_0} + \delta_0) \leq \frac{1}{2},$$

both cases imply

$$x \leq 2L(\sqrt{3h} + h), \qquad h < \delta_0.$$

In particular, by decreasing δ_0 if necessary, every such x lies in the interval $[0, r_0]$ from Assumption 3. Moreover,

$$|\partial_x b(y)| + |\partial_x \sigma(y)| \leq C_{\text{loc}}, \qquad y \in [0, r_0].$$

Now we sharpen this localization using the local boundedness of $\partial_x b$ and $\partial_x \sigma$ on $[0, r_0]$. Denote

$$\sigma_0 := \sigma(0), \qquad b_0 := b(0),$$

If $x - \sigma(x)\sqrt{3h} + b(x)h < 0$, then by the mean value theorem,

$$\begin{aligned} x &< \sigma(x)\sqrt{3h} - b(x)h \\ &= (\sigma_0 + x\partial_x\sigma(z_x))\sqrt{3h} - (b_0 + x\partial_x b(y_x))\, h \end{aligned} \tag{A.7}$$

for some $y_x, z_x \in [0, x]$. Therefore

$$x\left(1 - \partial_x\sigma(z_x)\sqrt{3h} + \partial_x b(y_x)h\right) < \sigma_0\sqrt{3h} - b_0 h.$$

After decreasing δ_0 if necessary, we have

$$1 - \partial_x\sigma(z_x)\sqrt{3h} + \partial_x b(y_x)h \geq \frac{1}{2}, \qquad h < \delta_0,$$

and hence

$$x < \sigma_0\sqrt{3h} + C_1 h,$$

where C_1 depends only on b_0, σ_0 and the local bound $C_{\rm loc}$.
Similarly, if $x - \sigma(x)\sqrt{3h} < 0$, then

$$x < \sigma(x)\sqrt{3h} = (\sigma_0 + x\partial_x\sigma(z_x))\sqrt{3h}, \tag{A.8}$$

so

$$x\left(1 - \partial_x\sigma(z_x)\sqrt{3h}\right) < \sigma_0\sqrt{3h}.$$

After decreasing δ_0 if necessary,

$$1 - \partial_x\sigma(z_x)\sqrt{3h} \geq \frac{1}{2},$$

and therefore

$$x < \sigma_0\sqrt{3h} + C_2 h,$$

where C_2 depends only on σ_0 and $C_{\rm loc}$. Let $C := C_1 \vee C_2$. We have shown that for every $h < \delta_0$,

$$\left\{x \geq 0 : \; x - \sigma(x)\sqrt{3h} < 0 \text{ or } x - \sigma(x)\sqrt{3h} + b(x)h < 0\right\} \subseteq [0, \sigma_0\sqrt{3h} + Ch].$$

Now choose $\delta \in (0, \delta_0]$ small enough so that

$$B_\delta := [0, \sigma_0\sqrt{3\delta} + C\delta] \subseteq [0, r_0].$$

Since b and σ are continuous on B_δ, define

$$b_1 := \min_{x\in B_\delta} b(x), \qquad b_2 := \max_{x\in B_\delta} b(x),$$

and

$$\sigma_1 := \min_{x\in B_\delta} \sigma(x), \qquad \sigma_2 := \max_{x\in B_\delta} \sigma(x).$$

By Assumption 3, after decreasing δ if necessary, $\sigma_1 > 0$. Therefore, for every $h < \delta$ and every x satisfying either boundary-layer inequality,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2],$$

as claimed. □

As a side remark, σ_i and b_i are different from $\sigma_0 = \sigma(0)$ and $b_0 = b(0)$ up to a term of order $\sqrt{h}$ because we have shown the boundary layer of our scheme is included in set B_δ. Lemma 4.1 is useful in the sense that it holds when we want to do error analysis for our scheme whenever we do not make a standard EM jump.

Lemma 4.2. There exists $\delta > 0$ such that $\forall h < \delta$, $\forall x$ such that $x - \sigma(x)\sqrt{3h} < 0$,

$$\lambda(x) \in [0, 1].$$

Proof. By Lemma 4.1, there exist $\delta_1 > 0$ and constants $b_1, b_2, \sigma_1, \sigma_2$, depending only on the local boundary constants and the linear-growth constant of b, σ, such that for every $h < \delta_1$ and every x with $x - \sigma(x)\sqrt{3h} < 0$, we have

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].$$

Hence

$$x < \sigma(x)\sqrt{3h} \leq \sigma_2\sqrt{3h}.$$

Therefore it suffices to prove the following algebraic statement: for all $(x, b, \sigma) \in [0, \sigma_2\sqrt{3h}] \times [b_1, b_2] \times [\sigma_1, \sigma_2]$, the quantity $\Lambda(x, b, \sigma)$ defined below (A.10) satisfies

$$0 \leq \Lambda(x, b, \sigma) \leq 1.$$

If $b_2 \leq 0$, choose $\delta_2 = 1$; if $b_2 > 0$, δ_2 is chosen in such a way that the following inequalities hold for all $h < \delta_2$,

$$\begin{aligned}\frac{\kappa}{\sqrt{3h}} &> \frac{2\kappa b_2 \sigma_2}{\sigma_1^2},\\ \sqrt{3h} &< \frac{\sigma_2}{2b_2},\end{aligned}$$

which implies that for all $b \in [b_1, b_2]$ and $\sigma \in [\sigma_1, \sigma_2]$

$$\begin{aligned}\frac{\kappa}{\sqrt{3h}} &> \frac{(2\kappa b - \sigma^2)\sigma_2}{\sigma^2},\\ \frac{\kappa\sigma}{\sqrt{3h}} &> -(\sigma^2 - 2\kappa b).\end{aligned} \tag{A.9}$$

Consider, $\forall h < \min\{\delta_1, \delta_2\}$,

$$\Lambda(x, b, \sigma) = \frac{(\sigma^2 - 2\kappa b)x^2 + 2\kappa\sigma^2 x + \sigma^4 h}{\left(\frac{\kappa\sigma}{\sqrt{3h}} + \sigma^2 - 2\kappa b\right)x^2 + \sigma^3\kappa\sqrt{3h} + \sigma^4 h}, \tag{A.10}$$

and

$$1 - \Lambda(x, b, \sigma) = \frac{\frac{\kappa\sigma}{\sqrt{3h}}\left(x - \sigma\sqrt{3h}\right)^2}{\left(\frac{\kappa\sigma}{\sqrt{3h}} + \sigma^2 - 2\kappa b\right)x^2 + \sigma^3\kappa\sqrt{3h} + \sigma^4 h}. \tag{A.11}$$

It suffices to conclude the lemma if we can show $\forall (x, b, \sigma) \in [0, \sigma_2\sqrt{3h}] \times [b_1, b_2] \times [\sigma_1, \sigma_2]$,

$$\Lambda(x, b, \sigma) \geq 0, \quad \text{and} \quad 1 - \Lambda(x, b, \sigma) \geq 0. \tag{A.12}$$

We consider three cases: (i) $b < \frac{\sigma^2}{2\kappa}$ (ii) $b > \frac{\sigma^2}{2\kappa}$; (iii) $b = \frac{\sigma^2}{2\kappa}$.

(1) For $b < \frac{\sigma^2}{2\kappa}$, we notice that both numerator (denote as $N_1(x, b)$) and denominator (denote as $D(x, b)$) of $\Lambda(x, b)$ are strictly positive. Besides, the numerator of $1 - \Lambda(x, b)$ is a square and its denominator is identical to $D(x, b)$, so it is strictly positive. (A.12) is verified.

(2) For $b > \frac{\sigma^2}{2\kappa}$, we have $2\kappa b - \sigma^2 > 0$, and in particular $b_2 > 0$. From the first inequality in (A.9), for every $x \in [0, \sigma_2\sqrt{3h}]$,

$$(2\kappa b - \sigma^2)x \leq (2\kappa b - \sigma^2)\sigma_2\sqrt{3h} < \kappa\sigma^2.$$

Hence

$$2\kappa\sigma^2 - (2\kappa b - \sigma^2)x > \kappa\sigma^2 > 0.$$

Using this, the numerator of Λ satisfies

$$\begin{aligned}N_1(x, b, \sigma) &= (\sigma^2 - 2\kappa b)x^2 + 2\kappa\sigma^2 x + \sigma^4 h\\ &= \sigma^4 h + x\big(2\kappa\sigma^2 - (2\kappa b - \sigma^2)x\big)\\ &> \sigma^4 h + \kappa\sigma^2 x\\ &> 0.\end{aligned}$$

For the denominator, by the second inequality in (A.9),

$$\frac{\kappa\sigma}{\sqrt{3h}} + \sigma^2 - 2\kappa b > 0.$$

Therefore

$$\begin{aligned} D(x,b,\sigma) &= \left(\frac{\kappa\sigma}{\sqrt{3h}} + \sigma^2 - 2\kappa b\right) x^2 + \sigma^3\kappa\sqrt{3h} + \sigma^4 h \\ &\geq \sigma^3\kappa\sqrt{3h} + \sigma^4 h \\ &> 0. \end{aligned}$$

Thus $\Lambda(x,b,\sigma) \geq 0$ on $[0, \sigma_2\sqrt{3h}] \times [b_1, b_2] \times [\sigma_1, \sigma_2]$. We also have $1 - \Lambda(x,b,\sigma) \geq 0$. Hence (A.12) holds in this case.

(3) For $b = \frac{\sigma^2}{2\kappa}$, by simple algebra,

$$\Lambda(x,b,\sigma) = \frac{2\sigma^2 x + 2b\sigma^2 h}{\frac{\sigma x^2}{\sqrt{3h}} + \sigma^3\sqrt{3h} + 2b\sigma^2 h} \geq 0,$$

$$1 - \Lambda(x,b,\sigma) = \frac{\frac{\sigma}{\sqrt{3h}}\left(x - \sigma\sqrt{3h}\right)^2}{\frac{\sigma x^2}{\sqrt{3h}} + \sigma^3\sqrt{3h} + 2b\sigma^2 h} \geq 0.$$

Hence (A.12) is verified in this case.

□

Lemma 4.7. There exists constant $C, \delta > 0$ dependent on $\sigma(x)$ and $b(x)$ such that $\forall h < \delta$ and $\forall x \in [0, \sigma_2\sqrt{3h}]$ such that $x - \sigma(x)\sqrt{3h} < 0$,

$$0 \leq \lambda(x) \leq C \cdot \left(\frac{x}{\sqrt{h}} + \sqrt{h}\right).$$

Proof. By Lemma 4.1, after decreasing δ if necessary, there exist constants $b_1, b_2, \sigma_1, \sigma_2$, independent of x and h, such that whenever $h < \delta$ and $x - \sigma(x)\sqrt{3h} < 0$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2],$$

and hence

$$0 \leq x < \sigma(x)\sqrt{3h} \leq \sigma_2\sqrt{3h}.$$

Let $B_M := |b_1| \vee |b_2|$, by Lemma 4.2, after decreasing δ if necessary, we already know that $\lambda(x) \in [0,1]$. It remains to prove the sharper upper bound. Write

$$\lambda(x) = \frac{N_h(x)}{D_h(x)},$$

where

$$N_h(x) := (\sigma^2(x) - 2\kappa b(x))x^2 + 2\kappa\sigma^2(x)x + \sigma^4(x)h,$$

and

$$D_h(x) := \left(\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x)\right) x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^4(x)h.$$

Decrease δ once more so that, for all $h < \delta$,

$$\frac{\kappa\sigma_1}{\sqrt{3h}} \geq 2\kappa B_M.$$

Then, for all x under consideration,

$$\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x) \geq \frac{\kappa\sigma_1}{\sqrt{3h}} - 2\kappa B_M \geq 0.$$

Therefore

$$D_h(x) \geq \sigma^3(x)\kappa\sqrt{3h} \geq \sigma_1^3\kappa\sqrt{3h}.$$

On the other hand,

$$|N_h(x)| \leq (\sigma_2^2 + 2\kappa B_M)x^2 + 2\kappa\sigma_2^2 x + \sigma_2^4 h.$$

Since $x \leq \sigma_2\sqrt{3h}$, we have $x^2 \leq 3\sigma_2^2 h$. Hence

$$\begin{aligned}\lambda(x) &\leq \frac{|N_h(x)|}{D_h(x)} \\ &\leq \frac{(\sigma_2^2 + 2\kappa B_M)\sigma_2^2 \cdot 3h + 2\kappa\sigma_2^2 x + \sigma_2^4 h}{\sigma_1^3\kappa\sqrt{3h}} \\ &= \frac{2\sigma_2^2}{\sqrt{3}\sigma_1^3} \cdot \frac{x}{\sqrt{h}} + \frac{4\sigma_2^4 + 6\kappa B_M\sigma_2^2}{\sqrt{3}\sigma_1^3\kappa} \cdot \sqrt{h}.\end{aligned} \tag{A.13}$$

Choosing

$$C := \max\left\{\frac{2\sigma_2^2}{\sqrt{3}\sigma_1^3}, \frac{4\sigma_2^4 + 6\kappa B_M\sigma_2^2}{\sqrt{3}\sigma_1^3\kappa}\right\}$$

completes the proof. □

Lemma 4.8. Let u be the solution of (4.3) satisfying corresponding boundary condition (4.2) and assume that $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$. Then, there exists δ small such that for all $h < \delta$ and $x \in \Omega_b$,

$$\partial_x u(t,x) = (x + M(x))\,\partial_{xx}u(t,x) + Re(x)$$

where

$$|Re(x)| \leq C(h + x) = \mathcal{O}(h + x),$$

and C depends on bounds of $\partial_x u, \partial_{xx}u, u^{(3)}$ and on the local bounds of b, σ and their derivatives up to order 2 on the boundary neighborhood $[0, r_0]$, but is independent of x and h.

Proof. Similar to proof of Lemma 3.2, apply Taylor expansion with remainder to (4.2) centered at x. By Lemma 4.1, after decreasing δ if necessary, every $x \in \Omega_b$ lies in $B_\delta \subset [0, r_0]$. Hence all intermediate points in the Taylor expansions below also lie in $[0, r_0]$, where the local coefficient bounds from Assumption 3 apply. We have

$$\begin{aligned}\sigma^2(0)\partial_x u(t,0) &= \sigma^2(x)\partial_x u(t,x) - x\sigma^2(x)\partial_{xx}u(t,x) - 2x\sigma(x)\sigma'(x)\partial_x u(t,x) \\ &\quad + \frac{x^2}{2}\partial_{xx}\left(\sigma^2(y_{1,x})\partial_x u(t,y_{1,x})\right) \\ &= \sigma^2(x)\partial_x u(t,x) - x\sigma^2(x)\partial_{xx}u(t,x) + \mathcal{O}(h + x)\end{aligned}$$

and

$$\begin{aligned}
&2\kappa b(0)\partial_x u(t,0) + \kappa\sigma^2(0)\partial_{xx}u(t,0) \\
&= 2\kappa\left(b\cdot\partial_x u - x\partial_x\left(b\cdot\partial_x u\right) + \frac{x^2}{2}\partial_{xx}\left(b\cdot\partial_x u\right)\right) \\
&\quad + \kappa\left(\sigma^2\cdot\partial_{xx}u - x\cdot\partial_x\left(\sigma^2\cdot\partial_{xx}u\right)\right) \\
&= 2\kappa b(x)\partial_x u(t,x) + \left(-2x\kappa b(x) + \kappa\sigma^2(x)\right)\partial_{xx}u(t,x) + \mathcal{O}(h+x)
\end{aligned}$$

where we omit function arguments in expansion above. We choose δ small such that $\sigma^2(x) - 2\kappa b(x) \neq 0$ for all $x \in [0, \sigma_2\sqrt{3h}]$ because we assume drift and diffusion terms to be continuous and $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$. This allows us to conclude that $(\sigma^2(x) - 2\kappa b(x))^{-1}$ to be bounded for $x \in [0, \sigma_2\sqrt{3h}]$. This allows us to finish proving the lemma by equating two expressions above and rearranging terms. □

Lemma 4.9. There exists $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_b$, we have,

$$\mathbb{E}_x(\Delta X_k) = \frac{\lambda(x)x^2}{2\sqrt{3h}\cdot\sigma(x)} + \frac{\sigma(x)\lambda(x)\sqrt{3h}}{2} + \lambda(x)b(x)h - x + \mathcal{O}(hx + h^2).$$

where

$$R_1(x) = \frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}} = \mathcal{O}(hx + h^2)$$

Proof. We first calculate $\mathbb{E}_x(X_{k+1})$ where $b(x) < 0$. According to (4.6), given $X_k = x$, X_{k+1} can take four different forms

(1) If $x + \sigma(x)Z_{k+1} \geq 0$ and $x + \sigma(x)Z_{k+1} + b(x)h \geq 0$, then

$$X_{k+1} = x + \sigma(x)Z_{k+1} + b(x)h \quad \text{where} \quad Z_{k+1} \geq -\frac{1}{\sigma(x)}\left(b(x)h + x\right)$$

(2) If $x + \sigma(x)Z_{k+1} \geq 0$ and $x + \sigma(x)Z_{k+1} + b(x)h \leq 0$, then

$$X_{k+1} = -x - \sigma(x)Z_{k+1} - b(x)h \quad \text{where} \quad -\frac{x}{\sigma(x)} \leq Z_{k+1} \leq -\frac{1}{\sigma(x)}\left(b(x)h + x\right)$$

(3) If $x + \sigma(x)Z_{k+1} \leq 0$ and $x + \sigma(x)Z_{k+1} + b(x)h \leq 0$, then

$$X_{k+1} = x + \sigma(x)Z_{k+1} - b(x)h \quad \text{where} \quad \frac{1}{\sigma(x)}\left(b(x)h - x\right) \leq Z_{k+1} \leq -\frac{x}{\sigma(x)}$$

(4) If $x + \sigma(x)Z_{k+1} \leq 0$ and $x + \sigma(x)Z_{k+1} + b(x)h \geq 0$, then

$$X_{k+1} = -x - \sigma(x)Z_{k+1} + b(x)h \quad \text{where} \quad Z_{k+1} \leq \frac{1}{\sigma(x)}\left(b(x)h - x\right)$$

Therefore,

$$
\begin{aligned}
\mathbb{E}_x(X_{k+1}) &= \lambda(x)\mathbb{E}_x\left(||x+\sigma(x)Z_{k+1}|+b(x)h|\right)+(1-\lambda(x))\cdot 0\\
&= \int_{-\frac{1}{\sigma(x)}(b(x)h+x)}^{\sqrt{3h}} (x+\sigma(x)y+b(x)h)\cdot\frac{\lambda(x)}{2\sqrt{3h}}\,dy\\
&\quad+\int_{-\frac{x}{\sigma(x)}}^{-\frac{1}{\sigma(x)}(b(x)h+x)} (-x-\sigma(x)y-b(x)h)\cdot\frac{\lambda(x)}{2\sqrt{3h}}\,dy\\
&\quad+\int_{\frac{1}{\sigma(x)}(b(x)h-x)}^{-\frac{x}{\sigma(x)}} (x+\sigma(x)y-b(x)h)\cdot\frac{\lambda(x)}{2\sqrt{3h}}\,dy\\
&\quad+\int_{-\sqrt{3h}}^{\frac{1}{\sigma(x)}(b(x)h-x)} (-x-\sigma(x)y+b(x)h)\cdot\frac{\lambda(x)}{2\sqrt{3h}}\,dy\\
&= \frac{\lambda(x)x^2}{2\sqrt{3h}\cdot\sigma(x)}+\frac{\sigma(x)\lambda(x)\sqrt{3h}}{2}+\lambda(x)b(x)h+\frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}}.
\end{aligned}
\tag{A.14}
$$

This gives that

$$
\mathbb{E}_x(\Delta X_k)=\frac{\lambda(x)x^2}{2\sqrt{3h}\cdot\sigma(x)}+\frac{\sigma(x)\lambda(x)\sqrt{3h}}{2}+\lambda(x)b(x)h-x+\frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}}. \tag{A.15}
$$

When the drift term $b(x)$ is positive calculation is simpler (as there is only absolute value on $x+\sigma(x)Z_{k+1}$), we can verify that:

$$
\mathbb{E}_x(\Delta X_k)=\frac{\lambda(x)x^2}{2\sqrt{3h}\cdot\sigma(x)}+\frac{\sigma(x)\lambda(x)\sqrt{3h}}{2}+\lambda(x)b(x)h-x. \tag{A.16}
$$

In other word, the proof is complete if the $b(x)\geq 0$ and $R_1=0$. Back to the case where $b(x)<0$, the first conditional moment of ΔX_k is

$$
\mathbb{E}_x(\Delta X_k)=(\text{A.16})+\frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}},
$$

and the remainder

$$
R_1(x)=\frac{\lambda(x)b(x)^2h^2}{\sigma(x)\sqrt{3h}}=\mathcal{O}(hx+h^2) \tag{A.17}
$$

because of Lemma 4.7 and the bounds $b(x)\in[b_1,b_2]$, $\sigma(x)\geq\sigma_1>0$ from Lemma 4.1. □

Lemma 4.10. There exists $\delta>0$ and $C>0$ such that $\forall h<\delta$ and $\forall x\in\Omega_b$, we have,

$$
\mathbb{E}_x(\Delta X_k^2)=-\frac{\lambda(x)}{\sigma(x)\sqrt{3h}}\cdot x^3+(1+\lambda(x))x^2-\lambda(x)\sigma(x)\sqrt{3h}\cdot x+\lambda(x)\sigma^2(x)\cdot h+R_2(x)
$$

where

$$
|R_2(x)|\leq C\cdot(hx+h^2)=\mathcal{O}(hx+h^2).
$$

Proof. Again, by Lemma 4.1, after decreasing δ if necessary, every $x\in\Omega_b$ lies in $B_\delta\subset[0,r_0]$. When drift is negative, recycle part of calculation in (A.14) and separate out remainder terms of order greater that 3/2, we have: (when drift is positive, $\frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}$ will be removed from $R_2(x)$, but we still reach same conclusion)

$$\begin{aligned}
&\mathbb{E}_x(\Delta X_k^2)\\
&=\lambda(x)\mathbb{E}_x\left((||x+\sigma(x)Z_{k+1}|+b(x)h|-x)^2\right)+(1-\lambda(x))x^2\\
&=\lambda(x)\mathbb{E}_x\left((|x+\sigma(x)Z_{k+1}|+b(x)h)^2-2x\,||x+\sigma(x)Z_{k+1}|+b(x)h|+x^2\right)+(1-\lambda(x))x^2\\
&=\lambda(x)\mathbb{E}_x\left(|x+\sigma(x)Z_{k+1}|^2-2x\,||x+\sigma(x)Z_{k+1}|+b(x)h|+x^2\right)+(1-\lambda(x))x^2+R_2'(x)\\
&=\lambda(x)\cdot\left(x^2+\sigma^2(x)h-2x\left(\frac{x^2}{2\sigma(x)\sqrt{3h}}+\frac{\sigma(x)\sqrt{3h}}{2}+b(x)h+\frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}\right)+x^2\right)\\
&\quad+(1-\lambda(x))x^2+R_2'(x)\\
&=\lambda(x)\cdot\left(2x^2+\sigma^2(x)h-2x\left(\frac{x^2}{2\sigma(x)\sqrt{3h}}+\frac{\sigma(x)\sqrt{3h}}{2}\right)\right)\\
&\quad+(1-\lambda(x))x^2-2x\lambda(x)\left(b(x)h+\frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}\right)+R_2'(x)\\
&=-\frac{\lambda(x)}{\sigma(x)\sqrt{3h}}\cdot x^3+(1+\lambda(x))x^2-\lambda(x)\sigma(x)\sqrt{3h}\cdot x+\lambda(x)\sigma^2(x)\cdot h+R_2(x)
\end{aligned}$$

where

$$R_2'(x)=\lambda(x)\mathbb{E}_x\left(b(x)^2h^2+2b(x)h\,|x+\sigma(x)Z_{k+1}|\right)$$

and

$$\begin{aligned}
R_2(x)&=-2x\lambda(x)\left(b(x)h+\frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}\right)+R_2'(x)\\
&=2x\lambda(x)\left(b(x)h+\frac{b(x)^2h^2}{\sigma(x)\sqrt{3h}}\right)+\lambda(x)\mathbb{E}_x\left(b(x)^2h^2+2b(x)h\,|x+\sigma(x)Z_{k+1}|\right).
\end{aligned}$$

We can bound $R_2(\cdot)$ using the local boundary-layer bounds from Lemma 4.1. Recall that $B_M:=|b_1|\vee|b_2|$ and $\sigma(x)\geq\sigma_1>0$ on the boundary layer:

$$|R_2(x)|\leq 2B_Mhx+\frac{2B_M^2h^2x}{\sigma_1\sqrt{3h}}+B_M^2h^2+2B_Mh\mathbb{E}_x\left(|x+\sigma(x)Z_{k+1}|\right)\lambda(x).$$

Hence, it suffices to complete the proof by showing that $h\mathbb{E}_x\left(|x+\sigma(x)Z_{k+1}|\right)\lambda(x)=\mathcal{O}(hx+h^2)$. By Lemma 4.7, and using $x\leq\sigma_2\sqrt{3h}$ and $|Z_{k+1}|\leq\sqrt{3h}$, we have

$$\begin{aligned}
h\mathbb{E}_x\left(|x+\sigma(x)Z_{k+1}|\right)|\lambda(x)|&\leq h\left(x+\sigma_2\sqrt{3h}\right)|\lambda(x)|\\
&\leq Ch^{3/2}\left(\frac{x}{\sqrt{h}}+\sqrt{h}\right)\\
&\leq C(hx+h^2).
\end{aligned}$$

□

Lemma 4.11. Assume that $b(0)\neq\frac{\sigma^2(0)}{2\kappa}$, there exist $\delta>0$ such that $\forall h<\delta$ and $\forall x\in\Omega_b$

$$2\left(\mathbb{E}_x(\Delta X_k)-hb(x)\right)\cdot(x+M(x))+\mathbb{E}_x(\Delta X_k^2)=\sigma^2(x)h+R_3(x)\tag{A.18}$$

where $|R_3(x)| \le C(hx+h^2) = \mathcal{O}(hx+h^2)$ for some constant C independent of x and h. The constant depends only on the boundary-layer bounds $b_1, b_2, \sigma_1, \sigma_2$, κ, and the lower bound of $|\sigma^2(x) - 2\kappa b(x)|$ in a sufficiently small neighborhood of 0.

Proof. Let

$$\mathbb{E}_x(\Delta X_k) = A_h(x) + R_1(x),$$

where $A_h(x)$ denotes the principal term from (A.16). By Lemma 4.7,

$$|R_1(x)| \le C_1(hx+h^2) \tag{A.19}$$

for some constant C_1 independent of h.

Likewise, by Lemma 4.10,

$$\mathbb{E}_x(\Delta X_k^2) = B_h(x) + R_2(x),$$

where

$$B_h(x) = -\frac{\lambda(x)}{\sigma(x)\sqrt{3h}}x^3 + (1+\lambda(x))x^2 - \lambda(x)\sigma(x)\sqrt{3h}\,x + \lambda(x)\sigma^2(x)h,$$

and

$$|R_2(x)| \le C_2(hx+h^2) \tag{A.20}$$

for some constant C_2 independent of h. We now substitute these expansions into the quantity of interest:

$$2\big(\mathbb{E}_x(\Delta X_k) - hb(x)\big)(x+M(x)) + \mathbb{E}_x(\Delta X_k^2),$$

where

$$M(x) = \frac{\kappa\sigma^2(x)}{\sigma^2(x) - 2\kappa b(x)}.$$

This gives

$$\begin{aligned} &2\big(\mathbb{E}_x(\Delta X_k) - hb(x)\big)(x+M(x)) + \mathbb{E}_x(\Delta X_k^2) \\ &\quad= \underbrace{2\big(A_h(x) - hb(x)\big)(x+M(x)) + B_h(x)}_{=:\mathcal{A}_h(x)} + 2R_1(x)(x+M(x)) + R_2(x). \end{aligned} \tag{A.21}$$

A direct rearrangement of $\mathcal{A}_h(x)$ yields

$$\begin{aligned} \mathcal{A}_h(x) = \lambda(x)&\left(\left(\frac{M(x)}{\sigma(x)\sqrt{3h}} + 1\right)x^2 + M(x)\sigma(x)\sqrt{3h} + \sigma^2(x)h + 2b(x)M(x)h\right) \\ &- x^2 - 2M(x)x - 2b(x)M(x)h + 2x(\lambda(x)-1)b(x)h. \end{aligned} \tag{A.22}$$

By the definition (4.7) of $\lambda(x)$, the bracketed terms are chosen precisely so that

$$\begin{aligned} &\lambda(x)\left(\left(\frac{M(x)}{\sigma(x)\sqrt{3h}} + 1\right)x^2 + M(x)\sigma(x)\sqrt{3h} + \sigma^2(x)h + 2b(x)M(x)h\right) \\ &\qquad\qquad - x^2 - 2M(x)x - 2b(x)M(x)h = \sigma^2(x)h. \end{aligned}$$

Hence

$$\mathcal{A}_h(x) = \sigma^2(x)h + 2x(\lambda(x)-1)b(x)h. \tag{A.23}$$

Substituting (A.23) into (A.21), we obtain

$$2\big(\mathbb{E}_x(\Delta X_k) - hb(x)\big)(x+M(x)) + \mathbb{E}_x(\Delta X_k^2) = \sigma^2(x)h + R_3(x),$$

where

$$R_3(x) := 2x(\lambda(x)-1)b(x)h + 2R_1(x)(x+M(x)) + R_2(x). \tag{A.24}$$

It remains to bound $R_3(x)$. Since $b(0) \neq \sigma^2(0)/(2\kappa)$, continuity implies that, for sufficiently small $\delta > 0$, the quantity

$$\sigma^2(x) - 2\kappa b(x)$$

is bounded away from 0 on $x \in [0, \sigma_2\sqrt{3h}]$ whenever $h < \delta$. Therefore $M(x)$ is uniformly bounded on this region. Using also the boundary-layer bound $b(x) \in [b_1, b_2]$, the fact that $|\lambda(x) - 1| \leq 1$, the remainder bounds (A.19)–(A.20), and $x \in [0, \sigma_2\sqrt{3h}]$, we conclude that

$$|R_3(x)| \leq C(hx + h^2)$$

for some constant C independent of h. This proves the claim. ☐

Lemma 4.12. Assume that $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, there exists $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_b$,

$$|\mathbb{E}_x(\Delta X_k)| \leq C(h + hx + h^2), \tag{A.25}$$

where $C > 0$ is independent of x and h and depends only on κ, the local boundary bounds for b, σ, and the nondegeneracy constant coming from $b(0) \neq \sigma^2(0)/(2\kappa)$.

Proof. By Lemma 4.1, after decreasing δ if necessary, all $x \in [0, \sigma_2\sqrt{3h}]$ lie in the fixed boundary neighborhood where the local coefficient bounds apply. Since

$$b(0) \neq \frac{\sigma^2(0)}{2\kappa},$$

continuity implies that, after decreasing δ if necessary, there exists $c_0 > 0$ such that

$$|\sigma^2(x) - 2\kappa b(x)| \geq c_0, \qquad x \in [0, \sigma_2\sqrt{3h}].$$

Hence $M(x)$ is uniformly bounded on this region. Moreover, since

$$M(0) = \frac{\kappa\sigma^2(0)}{\sigma^2(0) - 2\kappa b(0)} \neq 0,$$

continuity also implies that there exists $m_0 > 0$ such that

$$|M(x)| \geq m_0, \qquad x \in [0, \sigma_2\sqrt{3h}]$$

for all sufficiently small h. Decreasing δ once more so that

$$\sigma_2\sqrt{3h} \leq \frac{m_0}{2}, \qquad h < \delta,$$

we obtain

$$|x + M(x)| \geq |M(x)| - x \geq \frac{m_0}{2}, \qquad x \in [0, \sigma_2\sqrt{3h}].$$

We derive from remainder estimation in Lemma 4.11 and the crude bound $\mathbb{E}_x(\Delta X_k^2) \leq Ch$ that

$$\begin{aligned}
|\mathbb{E}_x(\Delta X_k)| &= \left| \frac{\sigma^2(x)h + R_3(x) - \mathbb{E}_x(\Delta X_k^2)}{2\,(x + M(x))} + hb(x) \right| \\
&\leq \frac{\sigma_2^2 h + |R_3(x)| + (\sigma_2\sqrt{3h} + B_M h)^2}{m_0} + B_M h \\
&\leq C(h + hx + h^2).
\end{aligned}$$

☐

Lemma 4.13. There exists $\delta > 0$ such that $\forall h < \delta$ and $\forall x \in \Omega_{br}$,

$$\left|\mathbb{E}_x(\Delta X_k^3)\right| \le K_4(hx + h^2), \tag{A.26}$$

where K_4 is a fixed constant dependent on drift, diffusion, and κ.

Proof. By Lemma 4.1, after decreasing δ if necessary, all $x \in [0, \sigma_2\sqrt{3h}]$ lie in the fixed boundary neighborhood where the local coefficient bounds apply. We begin with the case where $x - \sigma(x)\sqrt{3h} \le 0$.

Similar to proof of Lemma 3.8, fix $x \in [0, \sigma_2\sqrt{3h}]$, an increment ΔX_k is supported on $[-x, \sigma_2\sqrt{3h} + B_M h]$. We decompose ΔX_k into its positive and negative parts, so that $|\Delta X_k|^3 = (\Delta X_k^+)^3 + (\Delta X_k^-)^3$ and therefore

$$\mathbb{E}_x(|\Delta X_k|^3) = \mathbb{E}_x[(\Delta X_k^+)^3] + \mathbb{E}_x[(\Delta X_k^-)^3].$$

Positive part. Whenever $\Delta X_k > 0$ the increment comes from the continuous branch Z_k in Algorithm 2. This means the largest positive increment is $\sigma(x)\sqrt{3h} + b(x)h$, hence

$$\mathbb{E}_x[(\Delta X_k^+)^3] \le \lambda(x)\,(\sigma_2\sqrt{3h} + B_M h)^3. \tag{A.27}$$

Recall by Lemma 4.7, for $x \in [0, \sqrt{3h}]$, we have the following bound on $\lambda(x)$,

$$|\lambda(x)| \le \frac{2\sigma_2^2}{\sqrt{3}\sigma_1^3} \cdot \frac{x}{\sqrt{h}} + \frac{(4\sigma_2^4 + 6\kappa B_M \sigma_2^2)}{\sqrt{3}\sigma_1^3\kappa} \cdot \sqrt{h} \le S_1\left(\frac{x}{\sqrt{h}} + \sqrt{h}\right).$$

Combining with (A.27) gives

$$\begin{aligned}
\mathbb{E}_x[(\Delta X_k^+)^3] &\le S_1\left(\frac{x}{\sqrt{h}} + \sqrt{h}\right)(\sigma_2\sqrt{3h} + B_M h)^3 \\
&\le S_1\left(\sqrt{3}\sigma_2 x + B_M\sqrt{h}x + \sqrt{3}\sigma_2 h + B_M h^{3/2}\right)(\sigma_2\sqrt{3h} + B_M h)^2.
\end{aligned}$$

In other word, exist constant $K_4^+ > 0$ that

$$\mathbb{E}_x[(\Delta X_k^+)^3] \le K_4^+(hx + h^2). \tag{A.28}$$

Negative part. Because $\Delta X_k \ge -x$, we have $0 \le \Delta X_k^- \le x$, and therefore

$$(\Delta X_k^-)^3 \le x^3.$$

Since $x \in [0, \sigma_2\sqrt{3h}]$,

$$\mathbb{E}_x[(\Delta X_k^-)^3] \le x^3 \le 3\sigma_2^2 hx. \tag{A.29}$$

Combining (A.28) and (A.29) gives

$$\mathbb{E}_x(|\Delta X_k|^3) \;\le\; K_4^+ h^2 + (K_4^+ + 3\sigma_2^2)hx.$$

Since $|\mathbb{E}_x(\Delta X_k^3)| \le \mathbb{E}_x(|\Delta X_k|^3)$, the same bound holds for $|\mathbb{E}_x(\Delta X_k^3)|$ with $K_4 = K_4^+ + 3\sigma_2^2$. We have shown the lemma for $x \in \Omega_b$. It remains to consider $x \in \Omega_r$. In this region the scheme performs the reflected Euler–Maruyama update

$$X_{k+1} = \left|x + \sigma(x)Z_{k+1} + b(x)h\right|.$$

Let $\Delta X_k^{\mathrm{EM}} := \sigma(x)Z_{k+1} + b(x)h$ denote the unreflected Euler–Maruyama increment. By symmetry of Z_{k+1} and the boundary-layer bounds on b and σ,

$$\left|\mathbb{E}_x\left[\left(\Delta X_k^{\mathrm{EM}}\right)^3\right]\right| = \left|3b(x)\sigma^2(x)h^2 + b^3(x)h^3\right| \leq Ch^2.$$

The reflected and unreflected increments differ only on

$$A := \{x + \sigma(x)Z_{k+1} + b(x)h < 0\}.$$

Since $x \in \Omega_r$, Lemma 4.1 gives $x = \mathcal{O}(\sqrt{h})$, and the interval of Z_{k+1} values producing reflection has length $\mathcal{O}(h)$. Because Z_{k+1} is uniform on an interval of length $2\sqrt{3h}$, we have

$$\mathbb{P}_x(A) = \mathcal{O}(\sqrt{h}).$$

On A, both $|\Delta X_k|$ and $|\Delta X_k^{\mathrm{EM}}|$ are $\mathcal{O}(\sqrt{h})$, so

$$\left|\mathbb{E}_x\left[\Delta X_k^3 - \left(\Delta X_k^{\mathrm{EM}}\right)^3\right]\right| \leq Ch^{3/2}\mathbb{P}_x(A) = \mathcal{O}(h^2).$$

Hence

$$|\mathbb{E}_x(\Delta X_k^3)| \leq K_4' h^2 \leq K_4'(hx + h^2),$$

and the proof is complete.

□

Theorem 4.14. Given $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, there exists $\delta > 0$ and $C \geq 0$ depending on the relevant bounds of derivatives of u and on the local boundary constants for b, σ such that $\forall h < \delta$ and $\forall x \in \Omega_b$,

$$|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| \leq C(h^2 + hx) = \mathcal{O}(hx + h^2) \tag{A.30}$$

Proof. We choose δ small enough so that all lemmas we have proved in this section hold for $h < \delta$. In particular, by Lemma 4.1, for all $h < \delta$ and all $x \in \Omega_b$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].$$

These constants are local boundary-layer constants and are independent of x and h. Let u be the solution to the backward equation (4.3). We perform a Taylor expansion of u to 3rd order centered at (t_k, X_k). We write $u_k = u(t_k, X_k), \partial_x u_k = \partial_x u(t_k, X_k)$ (and similarly for $\partial_{xx} u_k, \partial_{xxx} u_k$), and we write Y_k for a point such that $Y_k \in [X_k, X_{k+1}]$ or $Y_k \in [X_{k+1}, X_k]$, and s_k will be a point in interval $s_k \in [kh, (k+1)h]$). Expanding gives

$$\begin{aligned} u_{k+1} - u_k = \partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2}\partial_{xx} u_k \cdot \Delta X_k^2 \\ + \frac{1}{2}\partial_{tt} u(s_k, X_k) \cdot h^2 + \frac{1}{6}\partial_{xxx} u(t_k, Y_k) \cdot \Delta X_k^3. \end{aligned} \tag{A.31}$$

We are interested in $\mathbb{E}_x(u_{k+1} - u_k)$ and know $x \in \Omega_b$. We now bound the expectation of each of the terms in the Taylor-expansion (A.31). We have

$$\mathbb{E}_x|\partial_{tt} u(s_k, X_k) \cdot h^2| \leq h^2 \|\partial_{tt} u\|_\infty$$
$$\mathbb{E}_x|\partial_{xxx} u(t_k, Y_k) \cdot \Delta X_k^3| \leq C(h^2 + hx)\|u^{(3)}\|_\infty,$$

where the second inequality follows from Lemma 4.13. We also have, using the backward equation (4.3) and Lemma 4.8,

$$
\begin{aligned}
&\mathbb{E}_x\left[\partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2}\partial_{xx} u_k \cdot \Delta X_k^2\right] \\
&\quad = \partial_x u(t_k,x)\left(\mathbb{E}_x(\Delta X_k) - b(x)h\right) + \frac{1}{2}\partial_{xx}u(t_k,x)\left(\mathbb{E}_x(\Delta X_k^2) - \sigma^2(x)h\right) \\
&\quad = \left[(x+M(x))\left(\mathbb{E}_x(\Delta X_k) - b(x)h\right) + \frac{1}{2}\left(\mathbb{E}_x(\Delta X_k^2) - \sigma^2(x)h\right)\right]\partial_{xx}u(t_k,x) \\
&\qquad + \left(\mathbb{E}_x(\Delta X_k) - b(x)h\right) Re(x) \\
&\quad = \frac{1}{2}\partial_{xx}u(t_k,x)R_3(x) + \left(\mathbb{E}_x(\Delta X_k) - b(x)h\right) Re(x).
\end{aligned}
\tag{A.32}
$$

The last step uses Lemma 4.11. Therefore, applying the triangle inequality to (A.31), and then using Lemma 4.12 and remainder estimates in Lemma 4.8, 4.11 shows that there exist constants M_1, M_2, independent of x, t, h, such that

$$
\begin{aligned}
|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| &\le \frac{1}{2}\|\partial_{xx}u\|_\infty \cdot |R_3(x)| + \left(|\mathbb{E}_x(\Delta X_k)| + B_M h\right)\cdot |Re(x)| \\
&\quad + h^2\|\partial_{tt}u\|_\infty + C(h^2 + hx)\|u^{(3)}\|_\infty \\
&\le M_1 h^2 + M_2 hx
\end{aligned}
$$

where $B_M := |b_1| \vee |b_2|$. □

Lemma 4.15. There exists $\delta > 0$ and constant C which depends only on the boundary-layer bounds such that $\forall h < \delta$ and $\forall x \in \Omega_r$,

$$|E_x(\Delta X_k) - b(x)h| \le C(hx + h^2) = \mathcal{O}(hx + h^2)$$

Proof. Recall that

$$\Omega_r := \left\{x \ge \sigma(x)\sqrt{3h} :\ x - \sigma(x)\sqrt{3h} + b(x)h \le 0\right\}.$$

For $x \in \Omega_r$ the scheme performs a reflected Euler–Maruyama step,

$$X_{k+1} = \left|x + \sigma(x)Z_{k+1} + b(x)h\right|, \qquad Z_{k+1} \sim \mathrm{Unif}[-\sqrt{3h}, \sqrt{3h}].$$

Hence

$$
\begin{aligned}
&\mathbb{E}_x(X_{k+1}) = \\
&\qquad \frac{1}{2\sqrt{3h}}\int_{-\frac{x+b(x)h}{\sigma(x)}}^{\sqrt{3h}} \left(x + \sigma(x)z + b(x)h\right)\, dz + \frac{1}{2\sqrt{3h}}\int_{-\sqrt{3h}}^{-\frac{x+b(x)h}{\sigma(x)}} \left(-x - \sigma(x)z - b(x)h\right)\, dz.
\end{aligned}
$$

A direct calculation gives

$$\mathbb{E}_x(X_{k+1}) = \frac{\sigma(x)\sqrt{3h}}{2} + \frac{(x+b(x)h)^2}{2\sigma(x)\sqrt{3h}}.$$

Therefore

$$\mathbb{E}_x(\Delta X_k) - b(x)h = \frac{\sigma(x)\sqrt{3h}}{2} + \frac{(x+b(x)h)^2}{2\sigma(x)\sqrt{3h}} - x - b(x)h$$
$$= \frac{\left(x+b(x)h-\sigma(x)\sqrt{3h}\right)^2}{2\sigma(x)\sqrt{3h}}.$$

In particular,

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| = \frac{\left(x+b(x)h-\sigma(x)\sqrt{3h}\right)^2}{2\sigma(x)\sqrt{3h}}.$$

Let $B_M := |b_1| \vee |b_2|$. By Lemma 4.1, after decreasing δ if needed, we have

$$\sigma(x) \geq \sigma_1 > 0, \qquad |b(x)| \leq B_M$$

for all $x \in \Omega_r$ and all $h < \delta$. Moreover, since $x \in \Omega_r$,

$$x \geq \sigma(x)\sqrt{3h}, \qquad x + b(x)h \leq \sigma(x)\sqrt{3h}.$$

Hence

$$0 \leq \sigma(x)\sqrt{3h} - x - b(x)h \leq -b(x)h \leq B_M h,$$

and therefore

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| \leq \frac{B_M^2 h^2}{2\sigma(x)\sqrt{3h}} \leq \frac{B_M^2}{2\sigma_1\sqrt{3}}\, h^{3/2}.$$

Finally, $x \in \Omega_r$ also implies

$$x \geq \sigma(x)\sqrt{3h} \geq \sigma_1\sqrt{3h},$$

so

$$h^{3/2} \leq \frac{1}{\sigma_1\sqrt{3}}\, hx.$$

Combining the last two bounds yields

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| \leq \frac{B_M^2}{6\sigma_1^2}\, hx \leq C(hx + h^2)$$

for some constant C independent of h and x. This completes the proof. □

Theorem 4.16. (Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in algorithm 2, situation 2)
Given $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, there exists $\delta > 0$ and $C \geq 0$ depending on the relevant derivative bounds of u and on the local boundary constants for b and σ such that $\forall h < \delta$ and $\forall x \in \Omega_r$,

$$|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| \leq C(h^2 + hx) = \mathcal{O}(hx + h^2) \tag{A.33}$$

Proof. The proof is a simplified version of proof of Theorem 4.14. By Lemma 4.1, after decreasing δ if necessary, for all $h < \delta$ and all $x \in \Omega_r$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].$$

Let $B_M := |b_1| \vee |b_2|$. Recall from regularity assumption and Lemma 4.13,

$$\mathbb{E}_x|\partial_{tt}u(s_k, X_k) \cdot h^2| \leq h^2 \|\partial_{tt}u\|_\infty$$
$$\mathbb{E}_x|\partial_{xxx}u(t_k, Y_k) \cdot \Delta X_k^3| \leq C(h^2 + hx)\|u^{(3)}\|_\infty,$$

We also have, using the fact that u solves the backward equation (4.3),

$$\begin{aligned}&\mathbb{E}_x\left[\partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2}\partial_{xx} u_k \cdot (\Delta X_k^2)\right]\\&\quad= \partial_x u(t_k, x) \cdot (\mathbb{E}_x(\Delta X_k) - b(x)h) + \frac{1}{2}\partial_{xx} u(t_k, x) \cdot \left(-\sigma^2(x) \cdot h + \mathbb{E}_x(\Delta X_k^2)\right).\end{aligned}$$

Apply Lemma 4.15 and apply the lower bound of $\mathbb{E}_x(X_{k+1})$ to the second conditional moment calculation, we have,

$$\begin{aligned}|E_x(\Delta X_k) - b(x)h| &\le C_2(hx + h^2)\\|E_x(\Delta X_k^2) - \sigma^2(x)h| &\le C_3(hx + h^2)\end{aligned}$$

Therefore, applying the triangle inequality to the Taylor expansion (A.31), and then using the first- and second-moment estimates above and Lemma 4.13, there exist constants M_1, M_2, independent of x, t, h, such that

$$\begin{aligned}|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| &\le \|\partial_x u\|_\infty \cdot C_2(hx + h^2) + \frac{1}{2}\|\partial_{xx} u\|_\infty \cdot C_3(hx + h^2)\\&\quad + h^2\|\partial_{tt} u\|_\infty + C_1(h^2 + hx)\|u^{(3)}\|_\infty\\&\le M_1 h^2 + M_2 hx.\end{aligned}$$

□

Lemma 4.17. When $b(0) = \frac{\sigma^2(0)}{2\kappa}$, there exist $\delta > 0$ and and constant C which depends only on the boundary-layer bounds such that $\forall h < \delta$ and $\forall x \in \Omega_b$,

$$|\mathbb{E}_x\left(\Delta X_k\right) - hb(x)| \le C(hx + h^2) = \mathcal{O}(hx + h^2).$$

Proof. As before,we know there exists $\delta > 0$ such that for all $h < \delta$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2], \qquad \lambda(x) \in [0, 1], \qquad \forall x \in \Omega_b.$$

It suffices to complete the proof if we show the inequality holds for all $x \in [0, \sigma_2\sqrt{3h}]$. Since $\sigma(0) > 0$, the assumption $b(0) = \sigma^2(0)/(2\kappa)$ implies $b(0) > 0$. By continuity, after further decreasing δ if necessary, $b(x) \ge 0$ for any $x \in \Omega_b$ and $\Omega_r = \emptyset$. We can check that when $b(x) = \frac{\sigma^2(x)}{2\kappa}$,

$$\mathbb{E}_x\left(\Delta X_k\right) = hb(x),$$

we shall expect that $=$ is replaced by $\approx$ within the boundary layer. We use Taylor expansion to make the argument more accurate, $\mathbb{E}_x\left(\Delta X_k\right)$ is a good approximation to $hb(x)$. We know that, denote $\sigma_0 := \sigma(0)$ and $b_0 := b(0)$,

$$\begin{aligned}\sigma^2(x) &= \left(\sigma_0 + x \cdot \partial_x \sigma(z_x)\right)^2,\\b(x) &= b_0 + x \cdot \partial_x b(y_x).\end{aligned}$$

Therefore,

$$\sigma^2(x) - 2\kappa b(x) = \sigma_0^2 - 2\kappa b_0 + \hat{R}(x) = \hat{R}(x) \tag{A.34}$$

where

$$|\hat{R}(x)| = \left|\left(2\sigma_0 \cdot \partial_x \sigma(z_x) - 2\kappa \cdot \partial_x b(y_x)\right) x + \left(\partial_x \sigma(z_x)\right)^2 \cdot x^2\right| \le \hat{K}(x + x^2).$$

where $\hat{K}$ depends only on the local first-derivative bounds of b and σ on the boundary neighborhood, and is independent of x and h. We substitute (A.34) into (4.7),

$$\lambda(x) = \frac{\hat{R}(x)x^2 + 2\kappa\sigma^2(x)x + \sigma^2(x)h \cdot \left(2\kappa b(x) + \hat{R}(x)\right)}{\frac{\kappa\sigma(x)}{\sqrt{3h}} \cdot x^2 + \hat{R}(x) \cdot x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^2(x)h \cdot \left(2\kappa b(x) + \hat{R}(x)\right)}$$

$$= \frac{\kappa\sigma^2(x)\left(2x + 2b(x)h + R(x)\right)}{\kappa\sigma^2(x)\left(\frac{x^2}{\sigma(x)\sqrt{3h}} + \sigma(x)\sqrt{3h} + 2b(x)h + R(x)\right)}$$
$$= \frac{2x + 2b(x)h + R(x)}{\frac{x^2}{\sigma(x)\sqrt{3h}} + \sigma(x)\sqrt{3h} + 2b(x)h + R(x)}$$

where

$$R(x) = \frac{\sigma^2(x)h + x^2}{\kappa\sigma^2(x)} \cdot \hat{R}(x)\,.$$

Recall $x \le \sigma_2\sqrt{3h}$,

$$\frac{\sigma^2(x)h + x^2}{\kappa\sigma^2(x)} \le \frac{\sigma_2^2 h + 3\sigma_2^2 h}{k\sigma_1^2} = Ch.$$

Hence,

$$|R(x)| \le Ch|\hat{R}(x)| \le Ch(x + x^2) \le C(hx + h^2).$$

Since the drift is positive, substitute $\lambda(x)$ back into (A.16), we have

$$\mathbb{E}_x(\Delta X_k) = \lambda(x) \cdot \left(\frac{x^2}{2\sigma(x)\sqrt{3h}} + \frac{1}{2}\sigma(x)\sqrt{3h} + b(x)h + \frac{1}{2}R(x) - \frac{1}{2}R(x)\right) - x$$
$$= x + b(x)h + \frac{1}{2}\left(1 - \lambda(x)\right)R(x) - x$$
$$= b(x)h + \frac{1}{2}\left(1 - \lambda(x)\right)R(x)$$

Hence,

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| = \left|\frac{1}{2}\left(1 - \lambda(x)\right)R(x)\right| \le C(hx + h^2)$$

because we know $\lambda(x) \in [0,1]$, $x \in [0, \sigma_2\sqrt{3h}]$, $|\hat{R}(x)| \le \hat{K}(x + x^2)$, and the coefficient $\frac{\sigma^2(x)h+x^2}{\kappa\sigma^2(x)}$ we scale $\hat{R}(\cdot)$ to $R(\cdot)$ is of order h using the boundary-layer bounds $\sigma_1 \le \sigma(x) \le \sigma_2$ from Lemma 4.1. □

Theorem 4.18. (Local Error of $\{X_k\}_{k\in\mathbb{N}}$ in algorithm 2, situation 3)
If $b(0) = \frac{\sigma^2(0)}{2\kappa}$, there exist $\delta > 0$ and $C \ge 0$, independent of x and h, depending only on the relevant bounds of derivatives of u and on the local boundary constants for b, σ such that $\forall h < \delta$ and for all $x \in \Omega_{br}$,

$$|\mathbb{E}\left(u_{k+1} - u_k | X_k = x\right)| \le C(h^2 + hx) = \mathcal{O}(hx + h^2). \tag{A.35}$$

Proof. The proof follows the same Taylor-expansion argument as in Theorem 4.14. Fix x such that $x - \sigma(x)\sqrt{3h} < 0$. By Lemma 4.1, after decreasing δ if necessary, for all $h < \delta$ and all x satisfying $x - \sigma(x)\sqrt{3h} < 0$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].$$

These constants are independent of x and h. By Taylor expansion of u centered at (t_k, x), with Y_k between x and X_{k+1} and $s_k \in [t_k, t_{k+1}]$, we have

$$\begin{aligned} u_{k+1} - u_k = \partial_t u_k\, h + \partial_x u_k\, \Delta X_k + \frac{1}{2}\partial_{xx} u_k\, \Delta X_k^2 \\ + \frac{1}{2}\partial_{tt} u(s_k, x)\, h^2 + \frac{1}{6}\partial_{xxx} u(t_k, Y_k)\, \Delta X_k^3. \end{aligned}$$

Taking conditional expectation given $X_k = x$ and using the backward equation (4.3), we obtain

$$\begin{aligned} &|\mathbb{E}\left(u_{k+1} - u_k \mid X_k = x\right)| \\ &\le |\partial_x u(t_k, x)| \cdot |\mathbb{E}_x(\Delta X_k) - b(x)h| + \frac{1}{2}\,|\partial_{xx} u(t_k, x)| \cdot \left|\mathbb{E}_x(\Delta X_k^2) - \sigma^2(x)h\right| \\ &\quad + \frac{h^2}{2}\|\partial_{tt} u\|_\infty + \frac{1}{6}\|u^{(3)}\|_\infty \left|\mathbb{E}_x(\Delta X_k^3)\right|. \end{aligned} \tag{A.36}$$

Because $b(0) = \frac{\sigma^2(0)}{2\kappa}$, the sticky boundary condition implies

$$\partial_{xx} u(t, 0) = 0.$$

Hence, by Taylor expansion in space,

$$\partial_{xx} u(t_k, x) = x\, u^{(3)}(t_k, y_x), \qquad \text{for some } y_x \in [0, x].$$

Therefore,

$$|\partial_{xx} u(t_k, x)| \le \|u^{(3)}\|_\infty\, x.$$

Next, by Lemma 4.17,

$$|\mathbb{E}_x(\Delta X_k) - b(x)h| \le C_1(hx + h^2).$$

Moreover, since $x - \sigma(x)\sqrt{3h} < 0$, the increment is supported in an interval of size $O(\sqrt{h})$, so

$$\mathbb{E}_x(\Delta X_k^2) \le C_2 h$$

for some constant $C_2 > 0$ independent of x and h because on the boundary layer the update either jumps to 0 or takes a reflected Euler–Maruyama step. Finally, by the third-moment bound already established for the boundary layer,

$$\left|\mathbb{E}_x(\Delta X_k^3)\right| \le C_3(hx + h^2).$$

Substituting these estimates into the inequality (A.36) yields

$$\begin{aligned} |\mathbb{E}\left(u_{k+1} - u_k \mid X_k = x\right)| &\le \|\partial_x u\|_\infty \cdot C_1(hx + h^2) + \frac{1}{2}\|u^{(3)}\|_\infty x \cdot \left(\sigma^2(x)h + C_2 h\right) \\ &\quad + \frac{h^2}{2}\|\partial_{tt} u\|_\infty + \frac{1}{6}\|u^{(3)}\|_\infty C_3(hx + h^2) \\ &\le M_1 h^2 + M_2 hx, \end{aligned}$$

for suitable constants $M_1, M_2 \ge 0$ independent of x and h, after possibly decreasing δ so that all preceding lemmas hold simultaneously. ∎

Proof of Theorem 4.3. The estimate on Ω_{br} follows from the three boundary-layer results already established. More precisely, when $b(0) \neq \frac{\sigma^2(0)}{2\kappa}$, Theorems 4.14 and 4.16 cover Ω_b and Ω_r, respectively. When $b(0) = \frac{\sigma^2(0)}{2\kappa}$, we have $b(0) > 0$, so by continuity, after possibly decreasing δ, one has $b(x) > 0$ throughout the boundary layer;

hence $\Omega_r = \varnothing$, and Theorem 4.18 yields the estimate on all of $\Omega_{br} = \Omega_b$. Therefore, after decreasing $\delta > 0$ finitely many times and enlarging the constant if needed, we may assume that

$$|\mathbb{E}\left(u_{k+1} - u_k \mid X_k = x\right)| \leq M_b(h^2 + hx), \qquad \forall x \in \Omega_{br},\ \forall h < \delta.$$

We now check $x \in \Omega_e$. Recall that, outside the boundary layer, the simulation scheme makes a standard Euler–Maruyama jump. Since

$$x - \sigma(x)\sqrt{3h} + b(x)h > 0,$$

the absolute value in (4.5) is inactive, and therefore

$$\Delta X_k = \sigma(x) Z_{k+1} + b(x)h.$$

Hence

$$\mathbb{E}_x(\Delta X_k) = b(x)h, \qquad \mathbb{E}_x(\Delta X_k^2) = \sigma^2(x)h + b^2(x)h^2,$$
$$\mathbb{E}_x(\Delta X_k^3) = 3b(x)\sigma^2(x)h^2 + b^3(x)h^3,$$

and

$$\begin{aligned} \mathbb{E}_x(\Delta X_k^4) &= \mathbb{E}_x\left(\sigma(x)Z_{k+1} + b(x)h\right)^4 \\ &= \frac{9}{5}\sigma^4(x)h^2 + 6b^2(x)\sigma^2(x)h^3 + b^4(x)h^4. \end{aligned} \tag{A.37}$$

The Taylor expansion upto fourth order is

$$\begin{aligned} u_{k+1} - u_k = \partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2}\partial_{xx} u_k \cdot \Delta X_k^2 \\ + \frac{1}{2}\partial_{tt} u(s_k, X_k) \cdot h^2 + \frac{1}{6}\partial_{xxx} u_k \cdot \Delta X_k^3 + \frac{1}{24}\partial_{xxxx} u(t_k, Y_k) \cdot \Delta X_k^4. \end{aligned} \tag{A.38}$$

Using the fact that u solves the backward equation (4.3), we know

$$\begin{aligned} &\mathbb{E}_x\left[\partial_t u_k \cdot h + \partial_x u_k \cdot \Delta X_k + \frac{1}{2}\partial_{xx} u_k \cdot (\Delta X_k^2)\right] \\ &= -b(x)\partial_x u(t_k, x) \cdot h - \frac{\sigma^2(x)}{2}\partial_{xx} u(t_k, x) \cdot h + \partial_x u_k \cdot b(x)h + \frac{1}{2}\partial_{xx} u(t_k, x) \cdot \left(\sigma^2(x)h + b^2(x)h^2\right) \\ &= \frac{1}{2} b^2(x)\partial_{xx} u(t_k, x)\, h^2. \end{aligned}$$

Taking conditional expectation in (A.38), and using the previous calculation, gives

$$\begin{aligned} |\mathbb{E}\left(u_{k+1} - u_k \mid X_k = x\right)| \leq{}& \frac{1}{2}|b(x)|^2 \|\partial_{xx} u\|_\infty h^2 + \frac{h^2}{2}\|\partial_{tt} u\|_\infty \\ &+ \frac{1}{6}\|u^{(3)}\|_\infty \left|3b(x)\sigma^2(x)h^2 + b^3(x)h^3\right| \\ &+ \frac{1}{24}\|u^{(4)}\|_\infty \left|\frac{9}{5}\sigma^4(x)h^2 + 6b^2(x)\sigma^2(x)h^3 + b^4(x)h^4\right|. \end{aligned}$$

By Assumption 4, there exists $L > 0$ such that

$$|b(x)| + |\sigma(x)| \leq L(1 + x), \qquad x \geq 0.$$

Hence, using $h<1$ and $(1+x)^4 \le C(1+x^4)$ for $x \ge 0$, we have

$$|b(x)|^2 + |b(x)|\,\sigma^2(x) + |b(x)|^3 h \le C(1+x^3) \le C(1+x^4),$$

and

$$\sigma^4(x) + b^2(x)\sigma^2(x)h + b^4(x)h^2 \le C(1+x^4).$$

Substituting these bounds and (A.37) into the previous estimate yields

$$|\mathbb{E}\,(u_{k+1} - u_k \mid X_k = x)| \le M_e h^2 (1+x^4), \qquad \forall x \in \Omega_e,\ \forall h < \delta,$$

for some constant M_e independent of x and h. □

Lemma 4.19. Denote $\tilde{\lambda}(\cdot)$ as the jump probability we propose in algorithm 1 and $\lambda(\cdot)$ as the one we propose in algorithm 2, if for the diffusion $\sigma(0)=1$ for a sticky diffusion, then

$$\left|\lambda(x) - \tilde{\lambda}(x)\right| \le K_{b,\sigma}\sqrt{h}, \quad \forall x \in [0, \sqrt{3h}]$$

where $K_{b,\sigma} > 0$ is a constant depending on the drift term $b(\cdot)$, diffusion $\sigma(\cdot)$, and sticky parameter κ.

Proof. Recall that

$$\lambda(x) = \frac{N_\sigma(x)}{D_\sigma(x)},$$

where

$$N_\sigma(x) := (\sigma^2(x) - 2\kappa b(x))x^2 + 2\kappa\sigma^2(x)x + \sigma^4(x)h$$

and

$$D_\sigma(x) := \left(\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x)\right)x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^4(x)h.$$

Similarly,

$$\tilde{\lambda}(x) = \frac{N_0(x)}{D_0(x)},$$

where

$$N_0(x) := x^2 + 2\kappa x + h, \qquad D_0(x) := \left(\frac{\kappa}{\sqrt{3h}} + 1\right)x^2 + h + \kappa\sqrt{3h}.$$

Using the same compactness argument as in Lemma 4.1, and using $\sigma(0)=1$, there exist $\delta_1 > 0$ and constants $b_1, b_2, \sigma_1, \sigma_2$, with $\sigma_1 > 0$, such that for every $h < \delta_1$ and every $x \in [0, \sqrt{3h}]$,

$$b(x) \in [b_1, b_2], \qquad \sigma(x) \in [\sigma_1, \sigma_2].$$

Set

$$B_M := |b_1| \vee |b_2|.$$

Moreover, since $\sigma(0)=1$ and σ is continuously differentiable near 0, after decreasing δ_1 if necessary, there exists a constant $C_\sigma > 0$, independent of x and h, such that for $n = 1,2,3,4$,

$$|\sigma^n(x) - 1| \le C_\sigma x \le C_\sigma\sqrt{3h}, \qquad \forall x \in [0, \sqrt{3h}], \quad h < \delta_1. \tag{A.39}$$

We also decrease δ_1 so that $\delta_1 \le 1$ and, for every $h < \delta_1$,

$$\frac{\kappa\sigma_1}{\sqrt{3h}} > \sigma_2^2 + 2\kappa B_M.$$

Then, for every $x \in [0, \sqrt{3h}]$,

$$D_\sigma(x) \geq \sigma^3(x)\kappa\sqrt{3h} \geq \kappa\sigma_1^3\sqrt{3h}. \tag{A.40}$$

Now define

$$\hat{\lambda}(x) := \frac{N_0(x)}{D_\sigma(x)}.$$

We first compare $\lambda(x)$ and $\hat{\lambda}(x)$. Since

$$N_\sigma(x) - N_0(x) = (\sigma^2(x) - 1)x^2 - 2\kappa b(x)x^2 + 2\kappa(\sigma^2(x) - 1)x + (\sigma^4(x) - 1)h,$$

using (A.39) and $x \leq \sqrt{3h}$ gives

$$|N_\sigma(x) - N_0(x)| \leq C_N h \tag{A.41}$$

for some constant $C_N > 0$ independent of x and h. Hence, by (A.40),

$$|\lambda(x) - \hat{\lambda}(x)| = \frac{|N_\sigma(x) - N_0(x)|}{D_\sigma(x)} \leq \frac{C_N h}{\kappa\sigma_1^3\sqrt{3h}} \leq C_N'\sqrt{h},$$

where $C_N' > 0$ is independent of x and h.

Next, we compare $\hat{\lambda}(x)$ and $\tilde{\lambda}(x)$. We have

$$D_\sigma(x) - D_0(x) = \left(\frac{\kappa(\sigma(x) - 1)}{\sqrt{3h}} + \sigma^2(x) - 1 - 2\kappa b(x)\right)x^2 + \kappa(\sigma^3(x) - 1)\sqrt{3h} + (\sigma^4(x) - 1)h.$$

Using (A.39) again, together with $x \leq \sqrt{3h}$ and $|b(x)| \leq B_M$, we obtain

$$|D_\sigma(x) - D_0(x)| \leq C_D h \tag{A.42}$$

for some constant $C_D > 0$ independent of x and h. Moreover,

$$\hat{\lambda}(x) - \tilde{\lambda}(x) = \frac{D_0(x) - D_\sigma(x)}{D_\sigma(x)} \cdot \frac{N_0(x)}{D_0(x)}.$$

Using Lemma 3.1, which gives $0 \leq \tilde{\lambda}(x) \leq 1$ on $[0, \sqrt{3h}]$, together with (A.40) and (A.42), we obtain

$$|\hat{\lambda}(x) - \tilde{\lambda}(x)| \leq \frac{|D_\sigma(x) - D_0(x)|}{D_\sigma(x)}|\tilde{\lambda}(x)| \leq \frac{C_D h}{\kappa\sigma_1^3\sqrt{3h}} \leq C_D'\sqrt{h},$$

where $C_D' > 0$ is independent of x and h.

Combining the two estimates, for every $h < \delta_1$ and every $x \in [0, \sqrt{3h}]$,

$$|\lambda(x) - \tilde{\lambda}(x)| \leq |\lambda(x) - \hat{\lambda}(x)| + |\hat{\lambda}(x) - \tilde{\lambda}(x)| \leq K_{b,\sigma}\sqrt{h}.$$

Here $K_{b,\sigma} := C_N' + C_D'$ depends only on κ, the local bounds $b_1, b_2, \sigma_1, \sigma_2$, and the local derivative bound for σ, but is independent of x and h. This completes the proof with $\delta = \delta_1$. $\square$

Lemma 4.4. There exists $\delta > 0$ and constant C such that for all $h < \delta$,

$$\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \cdot \mathbf{1}_{\Omega_{br}}(X_k)\right) < C$$

Proof. Throughout this proof we use the normalization $\sigma(0) = 1$ made before the statement of the lemma. By the proof of Lemma 4.1, there exist $\delta_B > 0$ and a constant $K_B > 0$, depending only on local bounds for b and σ near 0, such that for every $h < \delta_B$,

$$\Omega_{br} \subset B_h := [0, \sqrt{3h} + K_B h]. \tag{A.43}$$

Hence it is enough to prove a uniform bound with $\mathbf{1}_{B_h}$ in place of $\mathbf{1}_{\Omega_{br}}$. The rest of the proof is a discrete supersolution argument. We will construct a bounded barrier function $\varphi_h = e^{w_h}$ whose one-step drift is negative by an amount comparable to x on the enlarged boundary layer B_h. More precisely, the key estimate is

$$P\varphi_h(x) \leq \varphi_h(x)(1 - f_h(x) + C_1 h), \qquad f_h(x) \geq x\mathbf{1}_{B_h}(x).$$

Once this is proved, the time factor $e^{K_T(T-t_i)}$ will absorb the harmless $C_1 h$ error, giving a supersolution for the discrete occupation equation. A backward comparison then bounds the expected occupation sum.

We now modify the test function used in Lemma 3.5 so that it is unchanged near the boundary but becomes constant away from the boundary. This avoids requiring b and σ to be globally bounded. Fix $s > 16$ and $M > 0$. Choose a fixed number $R > 1$, independent of h. Later we decrease δ_B if necessary so that

$$B_h \subset [0, R/4].$$

Let $\chi \in C^2(\mathbb{R})$ satisfy

$$0 \leq \chi \leq 1, \qquad \chi(r) = 1 \text{ for } r \leq 0, \qquad \chi(r) = 0 \text{ for } r \geq 1.$$

Define

$$\eta_R(x) := \int_0^x \chi\left(\frac{y - R}{R}\right) dy.$$

Then $\eta_R(x) = x$ for $0 \leq x \leq R$, while $\eta_R(x) = A_R$ for $x \geq 2R$, where

$$A_R := \int_0^{2R} \chi\left(\frac{y - R}{R}\right) dy.$$

Moreover, $0 \leq \eta_R'(x) \leq 1$ and $|\eta_R''(x)| \leq C_R$ for a constant C_R depending only on R and the cutoff χ.

Define

$$w_h(x) := \begin{cases} M, & 0 \leq x \leq \sqrt{3h}/2, \\ M - s\left(\eta_R(x) - \frac{\sqrt{3h}}{2}\right), & x > \sqrt{3h}/2. \end{cases} \tag{A.44}$$

Then w_h agrees with the function used in Lemma 3.5 on $[0, R]$:

$$w_h(x) = \begin{cases} M, & 0 \leq x \leq \sqrt{3h}/2, \\ M - s\left(x - \frac{\sqrt{3h}}{2}\right), & \sqrt{3h}/2 < x \leq R. \end{cases}$$

Also, w_h is constant on $[2R, \infty)$. Let $\varphi_h(x) := e^{w_h(x)}$. Since $w_h(x) \leq M$ for all $x \geq 0$, $\varphi_h(x) \leq e^M$ uniformly in x and h. We first prove that there exist constants $C_1 > 0$ and $\delta_1 > 0$, independent of x and h, such that for all $h < \delta_1$,

$$P\varphi_h(x) \leq \varphi_h(x)\left(1 - f_h(x) + C_1 h\right), \qquad \forall x \geq 0, \tag{A.45}$$

where f_h satisfies

$$f_h(x) \geq x\mathbf{1}_{B_h}(x). \tag{A.46}$$

We introduce an auxiliary transition operator P_0. The role of P_0 is to separate the genuinely new diffusion effects from the SBM calculation. On B_h, the normalization $\sigma(0) = 1$ implies $\sigma(x) - 1 = O(\sqrt{h})$ and the scheme makes uniformly distributed jump of size $O(\sqrt{h})$, so the continuous update in the diffusion scheme differs from the SBM reflected update by only $O(h)$. The remaining complication is that P and P_0 correspond to different boundary layer . Given $Z \sim \mathcal{U}[-\sqrt{3h}, \sqrt{3h}]$,

$$P_0\varphi_h(x) = \begin{cases} \lambda(x)\mathbb{E}\left[\varphi_h(|x+Z|)\right] + (1-\lambda(x))\varphi_h(0), & 0 \le x \le \sqrt{3h}, \\ \mathbb{E}\left[\varphi_h(|x+Z|)\right], & x > \sqrt{3h}. \end{cases}$$

Thus P_0 is the same as the SBM transition operator, except that it uses the diffusion jump probability $\lambda(x)$ instead of $\tilde{\lambda}(x)$. We claim that for $x \in B_h$,

$$P\varphi_h(x) \le P_0\varphi_h(x) + C_2 h\varphi_h(x), \tag{A.47}$$

where C_2 is independent of x and h. We prove this by coupling the true diffusion update and the auxiliary update using the same random variable $Z \sim \mathcal{U}[-\sqrt{3h}, \sqrt{3h}]$.

First, after decreasing δ_1 if necessary, $B_h \subset [0, R/4]$. Since $\sigma(0) = 1$ and σ is locally Lipschitz near 0, there exists a constant $C_\sigma > 0$, independent of x and h, such that

$$|\sigma(x) - 1| \le C_\sigma x \le C_\sigma(\sqrt{3h} + K_B h) \le C\sqrt{h}, \qquad x \in B_h.$$

Similarly, b is locally bounded on a fixed compact neighborhood of 0, so

$$|b(x)| \le C, \qquad x \in B_h.$$

Thus, for all possible values of Z and all $x \in B_h$,

$$|x + Z| \le x + \sqrt{3h} \le C\sqrt{h},$$

and

$$||x + \sigma(x)Z| + b(x)h| \le x + |\sigma(x)Z| + |b(x)|h \le C\sqrt{h}.$$

After decreasing δ_1 once more so that $C\sqrt{h} < R$, all one-step positions appearing in the comparison below lie in $[0, R]$. We also record a bound on the Lipschitz constant of φ_h. Since $0 \le \eta_R' \le 1$, the function η_R is 1-Lipschitz. For h small enough that $\sqrt{3h}/2 < R$, we can write

$$w_h(x) = M - s\left(\eta_R(x) - \frac{\sqrt{3h}}{2}\right)_+,$$

Hence w_h is s-Lipschitz, uniformly in h. Since $w_h \le M$,

$$|\varphi_h(y) - \varphi_h(z)| = |e^{w_h(y)} - e^{w_h(z)}| \le e^M|w_h(y) - w_h(z)| \le se^M|y - z|. \tag{A.48}$$

Therefore φ_h is Lipschitz with constant $L_\varphi := se^M$, independent of x and h. Finally, on B_h we also have a uniform lower bound on $\varphi_h(x)$. Indeed, after decreasing δ_1 if necessary,

$$\varphi_h(x) = \exp(w_h(x)) \ge \exp\left(M - s\left(\frac{\sqrt{3h}}{2} + K_B h\right)\right) \ge e^{M/2}, \qquad x \in B_h. \tag{A.49}$$

Consequently, any estimate of the form Ch on B_h may be rewritten as $Ch\varphi_h(x)$ after increasing the constant C. We now compare the continuous branches. Let

$$Y_0 := |x + Z|$$

denote the continuous update in the auxiliary operator P_0. For the true diffusion scheme, define the continuous update

$$Y_1 := ||x + \sigma(x)Z| + b(x)h|$$

Using $||a| - |b|| \leq |a - b|$ gives

$$\begin{aligned}|Y_1 - Y_0| &= |||x + \sigma(x)Z| + b(x)h| - |x + Z|| \\ &\leq ||x + \sigma(x)Z| + b(x)h - |x + Z|| \\ &\leq ||x + \sigma(x)Z| - |x + Z|| + |b(x)|h \\ &\leq |(\sigma(x) - 1)Z| + |b(x)|h \\ &\leq Ch.\end{aligned} \tag{A.50}$$

Therefore,

$$|\varphi_h(Y_1) - \varphi_h(Y_0)| \leq L_\varphi |Y_1 - Y_0| \leq Ch \leq Ch\varphi_h(x). \tag{A.51}$$

It remains to control the possible mismatch between the two boundary thresholds $\sigma(x)\sqrt{3h}$ and $\sqrt{3h}$. A mismatch occurs only when one of the inequalities

$$x \leq \sigma(x)\sqrt{3h}, \qquad x \leq \sqrt{3h}$$

holds and the other fails. Equivalently, x lies between $\sqrt{3h}$ and $\sigma(x)\sqrt{3h}$. Hence

$$|x - \sigma(x)\sqrt{3h}| \leq |\sigma(x)\sqrt{3h} - \sqrt{3h}| = |\sigma(x) - 1|\sqrt{3h} \leq Ch. \tag{A.52}$$

In this mismatch region, we also need that $1 - \lambda(x)$ is of order h. Using the identity

$$1 - \lambda(x) = \frac{\frac{\kappa\sigma(x)}{\sqrt{3h}}\left(x - \sigma(x)\sqrt{3h}\right)^2}{\left(\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x)\right)x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^4(x)h}, \tag{A.53}$$

and using the same denominator lower-bound argument as in Lemma 4.19, after decreasing δ_1 if necessary, there exists $c > 0$ such that

$$\left(\frac{\kappa\sigma(x)}{\sqrt{3h}} + \sigma^2(x) - 2\kappa b(x)\right)x^2 + \sigma^3(x)\kappa\sqrt{3h} + \sigma^4(x)h \geq c\sqrt{h}, \qquad x \in B_h.$$

Combining this lower bound with (A.52) and (A.53), we obtain

$$1 - \lambda(x) \leq \frac{C\frac{1}{\sqrt{h}} \cdot h^2}{c\sqrt{h}} \leq Ch. \tag{A.54}$$

We now consider the possible threshold configurations. The following four cases only account for which operator regards x as being in the sticky boundary layer. If both operators make the same decision, the comparison follows from the $O(h)$ continuous-branch estimate. If their decisions differ, then x lies in the thin mismatch region, where $1 - \lambda(x) = O(h)$, so the extra sticky-jump contribution is also only $O(h)$.

(1) Suppose $x \in \Omega_b$ and $x \leq \sqrt{3h}$. Then

$$P\varphi_h(x) = \lambda(x)\mathbb{E}\left[\varphi_h(Y_1)\right] + (1 - \lambda(x))\varphi_h(0),$$

and

$$P_0\varphi_h(x) = \lambda(x)\mathbb{E}\left[\varphi_h(Y_0)\right] + (1 - \lambda(x))\varphi_h(0). \tag{A.55}$$

Thus, by (A.51)

$$P\varphi_h(x) - P_0\varphi_h(x) = \lambda(x)\mathbb{E}\left[\varphi_h(Y_1) - \varphi_h(Y_0)\right] \le Ch\varphi_h(x).$$

(2) Suppose that $x \in \Omega_b$ and $x > \sqrt{3h}$, then P_0 uses only the continuous branch,

$$\begin{aligned} P\varphi_h(x) - P_0\varphi_h(x) &= \lambda(x)\mathbb{E}\left[\varphi_h(Y_1) - \varphi_h(Y_0)\right] + (1-\lambda(x))\left(\varphi_h(0) - \mathbb{E}\left[\varphi_h(Y_0)\right]\right) \\ &\le Ch\varphi_h(x) + (1-\lambda(x)) \cdot C\varphi_h(x) \\ &\le Ch\varphi_h(x). \end{aligned}$$

where we use (A.51), (A.54) and the fact that

$$|\varphi_h(0) - \varphi_h(Y_0)| \le 2e^M \le 2e^{M/2}\varphi_h(x) \le C\varphi_h(x), \qquad x \in B_h,$$

which follows from the upper bound $\varphi_h \le e^M$ and the lower bound (A.49).

(3) Suppose that $x \notin \Omega_b$ and $x \le \sqrt{3h}$, then the true diffusion scheme uses only the continuous branch, while P_0 is in its sticky layer. Hence

$$\begin{aligned} P\varphi_h(x) - P_0\varphi_h(x) &= \mathbb{E}\left[\varphi_h(Y_1) - \varphi_h(Y_0)\right] + (1-\lambda(x))\left(\mathbb{E}\left[\varphi_h(Y_0)\right] - \varphi_h(0)\right) \\ &\le Ch\varphi_h(x) + (1-\lambda(x)) \cdot C\varphi_h(x) \\ &\le Ch\varphi_h(x). \end{aligned}$$

(4) Suppose that $x \notin \Omega_b$ and $x \ge \sqrt{3h}$, then both operators use only their continuous branch, and therefore

$$P\varphi_h(x) - P_0\varphi_h(x) = \mathbb{E}\left[\varphi_h(Y_1) - \varphi_h(Y_0)\right] \le Ch\varphi_h(x).$$

Taking C_2 large enough proves (A.47).

We now estimate $P_0\varphi_h$ on $[0, \sqrt{3h}]$. On this interval w_h agrees with the test function in Lemma 3.5. The only difference between P_0 and the SBM transition operator is the replacement of $\tilde{\lambda}(x)$ by $\lambda(x)$. Let $\tilde{P}_0$ denote the same auxiliary operator as P_0, but with λ replaced by $\tilde{\lambda}$. We wish to show $P_0\varphi_h(x) \approx \tilde{P}_0\varphi_h(x)$ so that we can reuse calculation done in Lemma 3.5. For $x \in [0, \sqrt{3h}]$, together with (A.55),

$$P_0\varphi_h(x) - \tilde{P}_0\varphi_h(x) = \left(\lambda(x) - \tilde{\lambda}(x)\right)\left(\mathbb{E}\left[\varphi_h(Y_0)\right] - \varphi_h(0)\right),$$

where $Y_0 = |x + Z| \le 2\sqrt{3h}$. Using the Lipschitz bound for φ_h (A.48) and the lower bound (A.49),

$$|\mathbb{E}\left[\varphi_h(Y_0)\right] - \varphi_h(0)| \le \mathbb{E}\left|\varphi_h(Y_0) - \varphi_h(0)\right| \le C\sqrt{h} \le C\sqrt{h}\,\varphi_h(x).$$

Apply Lemma 4.19,

$$|P_0\varphi_h(x) - \tilde{P}_0\varphi_h(x)| \le Ch\varphi_h(x), \qquad x \in [0, \sqrt{3h}].$$

We now estimate $P_0\varphi_h(x) - \varphi_h(x)$ by the case-by-case calculation in Lemma 3.5.

(1) For $x \in [0, \sqrt{3h}/2]$,

$$P_0\varphi_h(x) \le \varphi_h(x)\left(1 - f_h(x) + C_3 h\right),$$

where

$$f_h(x) := \frac{s}{4} \cdot \frac{\tilde{\lambda}(x)}{2}\sqrt{3h} + s \cdot \frac{\tilde{\lambda}(x)x^2}{2\sqrt{3h}}.$$

The proof of Lemma 3.5 shows that, for $s > 16$,

$$f_h(x) \ge x, \qquad x \in [0, \sqrt{3h}/2].$$

(2) Likewise, for $x \in [\sqrt{3h}/2, \sqrt{3h}]$,

$$P_0\varphi_h(x) \le \varphi_h(x)\left(1 - f_h(x) + C_3 h\right),$$

where

$$f_h(x) := \frac{s\tilde{\lambda}(x)}{16}\sqrt{3h} + \frac{s\tilde{\lambda}(x)x^2}{4\sqrt{3h}} + \frac{s\tilde{\lambda}(x)x}{4} + \frac{s\sqrt{3h}}{2} - sx.$$

Again, the proof of Lemma 3.5 shows that, for $s > 16$,

$$f_h(x) \ge x, \qquad x \in [\sqrt{3h}/2, \sqrt{3h}].$$

(3) The preceding SBM comparison covers $[0, \sqrt{3h}]$. The enlarged set B_h contains one additional strip of width $O(h)$, namely $[\sqrt{3h}, \sqrt{3h} + K_B h]$. This strip appears only because the diffusion boundary layer is not exactly the SBM boundary layer, so it must be checked directly. It remains to treat the thin strip $[\sqrt{3h}, \sqrt{3h} + K_B h]$. On this interval P_0 performs standard EM jump without reflection because $x \ge \sqrt{3h}$. Also $x + Z \in [0, R]$ for h sufficiently small, so w_h agrees with the original function used in Lemma 3.5. A direct calculation gives, uniformly for $x \in [\sqrt{3h}, \sqrt{3h} + K_B h]$,

$$\begin{aligned} P_0\varphi_h(x) - \varphi_h(x) &= \left(\frac{3}{4} - \frac{x}{2\sqrt{3h}}\right)e^M + \frac{e^M}{2\sqrt{3h}}\int_{\sqrt{3h}/2}^{\sqrt{3h}+x} e^{-s(z-\sqrt{3h}/2)}\,dz - e^{w_h(x)} \\ &= -e^{w_h(x)}\left[\frac{s}{4}\left(\frac{x^2}{\sqrt{3h}} - 3x + \frac{9}{4}\sqrt{3h}\right) + \mathcal{O}(h)\right]. \end{aligned} \tag{A.56}$$

Thus on this thin strip define

$$f_h(x) := \frac{s}{4}\left(\frac{x^2}{\sqrt{3h}} - 3x + \frac{9}{4}\sqrt{3h}\right).$$

For $x \in [\sqrt{3h}, \sqrt{3h} + K_B h]$,

$$f_h'(x) = \frac{s}{4}\left(\frac{2x}{\sqrt{3h}} - 3\right) \le -\frac{s}{8}$$

after decreasing δ_1 so that $2K_B h/\sqrt{3h} \le 1/2$. Hence f_h is decreasing on the thin strip. Therefore

$$\begin{aligned} f_h(x) &\ge f_h(\sqrt{3h} + K_B h) \\ &= \frac{s}{16}\sqrt{3h} - \frac{sK_B}{4}h + \frac{sK_B^2}{4}\frac{h^2}{\sqrt{3h}}. \end{aligned} \tag{A.57}$$

Since $s > 16$, after decreasing δ_1 if necessary,

$$f_h(\sqrt{3h} + K_B h) > \sqrt{3h} + K_B h.$$

Hence

$$f_h(x) \ge x, \qquad x \in [\sqrt{3h}, \sqrt{3h} + K_B h].$$

Combining the estimates for $P_0\varphi_h(x) - \varphi_h(x)$ on B_h with (A.47), we obtain

$$P\varphi_h(x) \le \varphi_h(x)(1 - f_h(x) + Ch), \qquad x \in B_h.$$

We now prove the corresponding estimate outside B_h. For $x \notin B_h$, set

$$f_h(x) := 0.$$

Since $\Omega_{br} \subset B_h$, if $x \notin B_h$, then the true scheme is outside the boundary layer and the standard Euler–Maruyama update is nonnegative for every possible Z. Thus

$$X_{k+1} = x + \sigma(x)Z + b(x)h.$$

By Assumption 4, there exists $L > 0$ such that

$$|b(x)| + |\sigma(x)| \le L(1 + x), \qquad x \ge 0.$$

If we can assume global boundedness on drift and diffusion, this part is straightforward because the usual Taylor expansion gives an order-h one-step bound for the test function under bounded coefficients. Without global boundedness assumption, we analyze $P\varphi_h(x)$ in

$$B_h^c \cap [0, R/2], \qquad [R/2, 4R], \qquad [4R, \infty).$$

The first region exploits the piecewise linear structure of $w(x)$ and boundedness of $\sigma(x)$, $b(x)$ in the compact domain. The second region exploits the smoothness of φ and the boundedness of drift and diffusion. The third region, φ is constant by construction and we can make sure the next step is within the constant region by linear growth assumption.

(1) Consider $x \in B_h^c \cap [0, R/2]$. On $[0, R]$, b and σ are bounded. For h sufficiently small, every possible value of $x + \sigma(x)Z + b(x)h$ lies in $[0, R]$. Since $x \notin B_h$, we also have $x > \sqrt{3h}/2$. On $[0, R]$, the function w_h is flat-linear, and for $y \in [0, R]$ and $x > \sqrt{3h}/2$,

$$w_h(y) - w_h(x) \le -s(y - x).$$

Therefore, by Taylor expansion around 0, using first and second moment of uniform random variable and the boundedness of σ, b in $[0, R]$, there exists C_4 is independent of x and h such that

$$\frac{P\varphi_h(x)}{\varphi_h(x)} \le \mathbb{E}\left[e^{-s(\sigma(x)Z + b(x)h)}\right] \le 1 + C_4 h.$$

(2) Consider $x \in [R/2, 4R]$. For h sufficiently small, every possible value of

$$Y := x + \sigma(x)Z + b(x)h$$

lies in $[R/4, 5R]$. Also $\sqrt{3h}/2 < R/4$, so φ_h is C^2 on $[R/4, 5R]$. There exist constants $c_\varphi, C_\varphi > 0$, depending on s, M, R and χ but not on x or h, such that

$$0 < c_\varphi \le \varphi_h(y), \qquad |\varphi_h'(y)| + |\varphi_h''(y)| \le C_\varphi, \qquad y \in [R/4, 5R].$$

Taylor expansion gives

$$\varphi_h(Y) = \varphi_h(x) + \varphi_h'(x)(Y - x) + \frac{1}{2}\varphi_h''(\xi)(Y - x)^2$$

for some ξ between x and Y. Taking expectations and using $\mathbb{E}Z = 0$ and $\mathbb{E}Z^2 = h$,

$$|P\varphi_h(x) - \varphi_h(x)| \le Ch.$$

Since $\varphi_h(x) \ge c_\varphi$ on this compact interval, this implies

$$P\varphi_h(x) \le \varphi_h(x)(1 + C_5 h), \qquad x \in [R/2, 4R],$$

where C_5 is independent of x and h.

(3) Consider $x \geq 4R$. By the linear-growth assumption, for all possible Z,

$$x + \sigma(x)Z + b(x)h \geq x - L(1+x)(\sqrt{3h} + h).$$

Since $R \geq 1$, $1 + x \leq \frac{5}{4}x$ for $x \geq 4R$. After decreasing δ_1 so that

$$\frac{5}{4}L(\sqrt{3h} + h) \leq \frac{1}{2},$$

we get

$$x + \sigma(x)Z + b(x)h \geq \frac{x}{2} \geq 2R.$$

Thus both x and X_{k+1} lie in the region where w_h is constant. Therefore

$$P\varphi_h(x) = \varphi_h(x), \qquad x \geq 4R.$$

Taking C_1 to be larger than the constants appearing in the estimates above, and decreasing δ_1 if necessary, we obtain (A.45) for all $x \geq 0$ and all $h < \delta_1$. Moreover, by construction,

$$f_h(x) \geq x\mathbf{1}_{B_h}(x).$$

Since f_h is supported on B_h and the formulas above are all of order $\sqrt{h}$, there exists $C_6 > 0$, independent of x and h, such that

$$0 \leq f_h(x) \leq C_6\sqrt{h}, \qquad \forall x \geq 0, \quad h < \delta_1. \tag{A.58}$$

We now introduce the time-dependent supersolution

$$V(t_i, x) := e^{K_T(T-t_i)}\varphi_h(x), \qquad i = 0, \ldots, N. \tag{A.59}$$

The constant $K_T > 0$ will be chosen below. Using (A.45),

$$\begin{aligned} PV(t_i, x) - V(t_i, x) &= e^{K_T(T-t_i-h)}P\varphi_h(x) - e^{K_T(T-t_i)}\varphi_h(x) \\ &\leq e^{K_T(T-t_i)}\varphi_h(x)\left[e^{-K_T h}(1 - f_h(x) + C_1 h) - 1\right]. \end{aligned} \tag{A.60}$$

After decreasing δ_1 if necessary,

$$e^{-K_T h} = 1 - K_T h + r_{K_T}(h), \qquad |r_{K_T}(h)| \leq C_7(K_T)h^2.$$

Using this expansion in (A.60), together with (A.58), gives

$$PV(t_i, x) - V(t_i, x) \leq -e^{K_T(T-t_i)}\varphi_h(x)\left(f_h(x) + (K_T - C_1)h - C_8(K_T)h^{3/2}\right).$$

Choose

$$K_T := C_1 + 1.$$

After decreasing δ_1 again, we may assume

$$(K_T - C_1)h - C_8(K_T)h^{3/2} \geq 0.$$

Hence

$$PV(t_i, x) - V(t_i, x) \leq -e^{K_T(T-t_i)}\varphi_h(x)f_h(x). \tag{A.61}$$

Finally, decrease δ_1 once more so that

$$M - s\left(\frac{\sqrt{3h}}{2} + K_B h\right) \geq 0.$$

Then $w_h(x) \geq 0$ on B_h, so $e^{K_T(T-t_i)}\varphi_h(x) \geq 1$ on B_h. Combining this with (A.46) and (A.61), we conclude that

$$PV(t_i, x) - V(t_i, x) \leq -x\mathbf{1}_{B_h}(x), \qquad i = 0, \ldots, N-1, \quad x \geq 0. \tag{A.62}$$

Define

$$U_B(t_i, x) := \mathbb{E}\left(\sum_{k=i}^{N-1} X_k \mathbf{1}_{B_h}(X_k) \,\middle|\, X_i = x\right).$$

By Lemma 3.9, applied with $g(t_i, x) = x\mathbf{1}_{B_h}(x)$, we have

$$PU_B(t_i, x) - U_B(t_i, x) = -x\mathbf{1}_{B_h}(x), \qquad i = 0, \ldots, N-1,$$

with terminal condition $U_B(T, x) = 0$. On the other hand, $V(T, x) = \varphi_h(x) \geq 0 = U_B(T, x)$, and (A.62) shows that V is a supersolution. A backward induction, exactly as in the proof of Lemma 3.5, gives

$$U_B(t_i, x) \leq V(t_i, x), \qquad i = 0, \ldots, N, \quad x \geq 0.$$

Since $w_h(x) \leq M$ for all $x \geq 0$,

$$V(t_i, x) \leq e^{K_T T + M}, \qquad i = 0, \ldots, N, \quad x \geq 0.$$

Therefore

$$U_B(t_0, x) \leq e^{K_T T + M}, \qquad x \geq 0.$$

Using $\Omega_{br} \subset B_h$, we get

$$\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \mathbf{1}_{\Omega_{br}}(X_k) \,\middle|\, X_0 = x\right) \leq U_B(t_0, x) \leq e^{K_T T + M}.$$

If X_0 is random, taking expectation and using the tower property gives

$$\mathbb{E}\left(\sum_{k=0}^{N-1} X_k \mathbf{1}_{\Omega_{br}}(X_k)\right) \leq e^{K_T T + M}.$$

Thus the desired bound holds with $C = e^{K_T T + M}$, which is independent of h. □

Proof of Lemma 4.5. It suffices to prove the one-step estimate

$$\mathbb{E}_x(X_{k+1}^4) \leq (1 + Ch)x^4 + Ch,$$

for a constant C independent of x and h.

First consider $x \in \Omega_e$. In this region the scheme makes a standard Euler–Maruyama jump, and the absolute value is inactive, so

$$X_{k+1} = x + b(x)h + \sigma(x)Z_{k+1}, \qquad Z_{k+1} \sim \mathrm{Unif}[-\sqrt{3h}, \sqrt{3h}].$$

Using $\mathbb{E}Z_{k+1} = 0$, $\mathbb{E}Z_{k+1}^3 = 0$, $\mathbb{E}Z_{k+1}^2 = h$, and $\mathbb{E}Z_{k+1}^4 = \frac{9}{5}h^2$, we have

$$\begin{aligned}\mathbb{E}_x(X_{k+1}^4) &= (x+b(x)h)^4 + 6(x+b(x)h)^2\sigma^2(x)h + \frac{9}{5}\sigma^4(x)h^2 \\ &= x^4 + 4x^3b(x)h + 6x^2b^2(x)h^2 + 4xb^3(x)h^3 + b^4(x)h^4 \\ &\quad + 6x^2\sigma^2(x)h + 12xb(x)\sigma^2(x)h^2 + 6b^2(x)\sigma^2(x)h^3 + \frac{9}{5}\sigma^4(x)h^2.\end{aligned} \tag{A.63}$$

By the linear-growth assumption, $|b(x)| + |\sigma(x)| \le L(1+x)$. Hence, using $h < 1$ and $(1+x)^4 \le C(1+x^4)$ for $x \ge 0$, each term after x^4 is bounded by $Ch(1+x^4)$. For example,

$$|x^3b(x)h| \le Chx^3(1+x) \le 2Ch(1+x^4),$$

$$x^2b^2(x)h^2 + x^2\sigma^2(x)h \le Chx^2(1+x)^2 \le 3Ch(1+x^4),$$

and the remaining terms are bounded similarly by $Ch(1+x)^4 \le Ch(1+x^4)$. Therefore,

$$\mathbb{E}_x(X_{k+1}^4) \le x^4 + Ch(1+x^4) \le (1+Ch)x^4 + Ch, \qquad x \in \Omega_e.$$

Next consider $x \in \Omega_{br}$. By Lemma 4.1, after decreasing δ if necessary, $x \le C\sqrt{h}$ and b, σ are bounded on the boundary layer. The scheme either jumps to 0 or makes a reflected Euler–Maruyama-type update, so in either case

$$X_{k+1} \le x + \sigma_2\sqrt{3h} + B_M h \le C\sqrt{h}.$$

Hence

$$\mathbb{E}_x(X_{k+1}^4) \le Ch^2 \le Ch \le (1+Ch)x^4 + Ch, \qquad x \in \Omega_{br}.$$

Combining the two cases gives

$$\mathbb{E}_x(X_{k+1}^4) \le (1+Ch)x^4 + Ch, \qquad x \ge 0.$$

Taking expectation with respect to X_k yields

$$\mathbb{E}(X_{k+1}^4) \le (1+Ch)\mathbb{E}(X_k^4) + Ch.$$

By the discrete Gronwall inequality, for $0 \le k \le N$ and $Nh = T$,

$$\mathbb{E}(X_k^4) \le e^{CT}\left(\mathbb{E}(X_0^4) + CT\right).$$

Therefore,

$$\sup_{0\le k\le N} \mathbb{E}(1+X_k^4) \le C_T,$$

where C_T is independent of h. □